

Modern Visualization Topics

Visualizing DL (explanation), Immersive Viz (inhabitation), Multimodal Viz (integration).

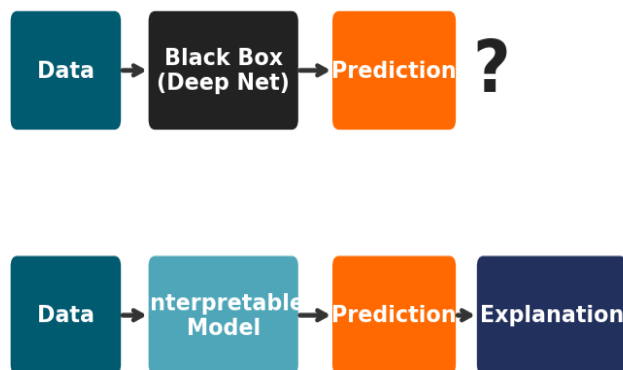
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Tandon School of Engineering, NYU

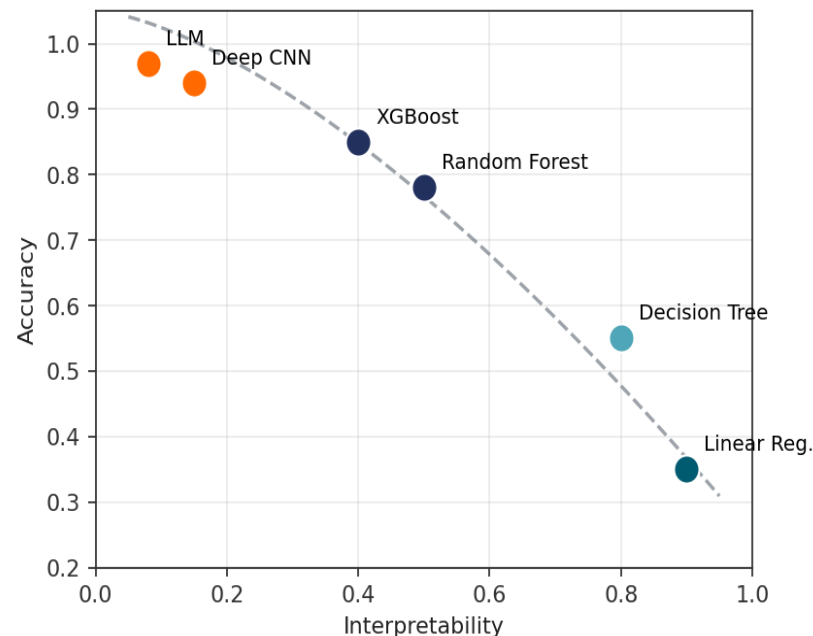
Information Visualization

From Black Box to Visible Model — A Viz Task

Black Box vs. Interpretable Pipeline



Interpretability-Accuracy Trade-off



"Explanation is not a model property — it is a visualization we construct around the model."

THE VIZ TASK

Take $f(x)$ — a function of millions of parameters — and render a picture a human can read. What data? What task? What idiom? (Munzner what-why-how)

DESIGN SPACE

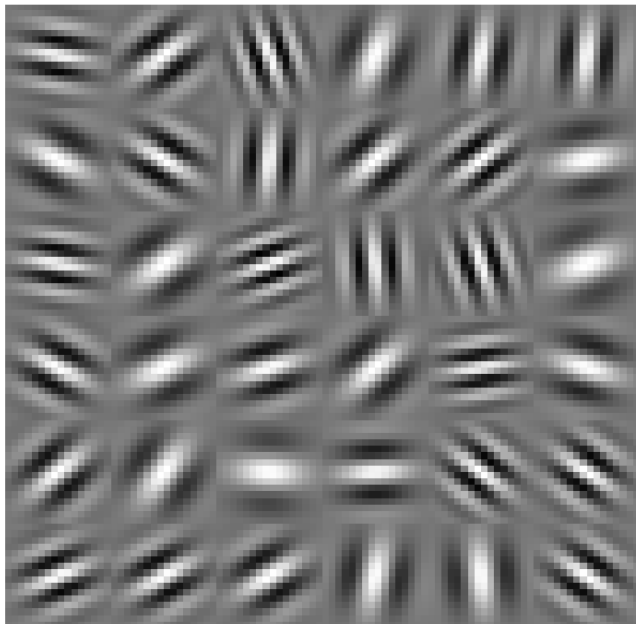
Attribution viz (WHERE it looked).
 Behavior viz (WHEN / UNDER WHAT).
 Structure viz (WHAT it encodes).
 Three families, three idiom sets.

CORE PRINCIPLE

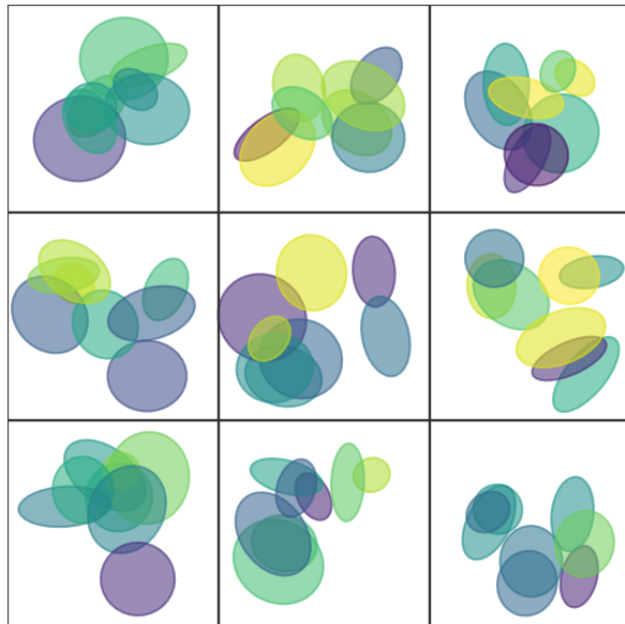
Faithful + legible. A correct viz the reader cannot parse is no viz.
 Model interpretation is ultimately a visualization design problem.

CNN Filters — The Small-Multiples Idiom

Layer 1: Edge & Color Detectors



Layer 5: Object-level Features



Python

```
# matplotlib small-multiples grid
fig, ax = plt.subplots(8, 8, figsize=(8, 8))
for i, a in enumerate(ax.flat):
    w = weights[i].mean(0)
    vmax = np.abs(w).max() # per-filter normalize
    a.imshow(w, cmap='RdBu_r', vmin=-vmax, vmax=vmax); a.axis('off')
```

ID I O M

Small multiples (Tufte 1983):
same encoding, repeated, so the eye
compares at a glance. The grid IS
the visualization.

ENCODING CHOICES

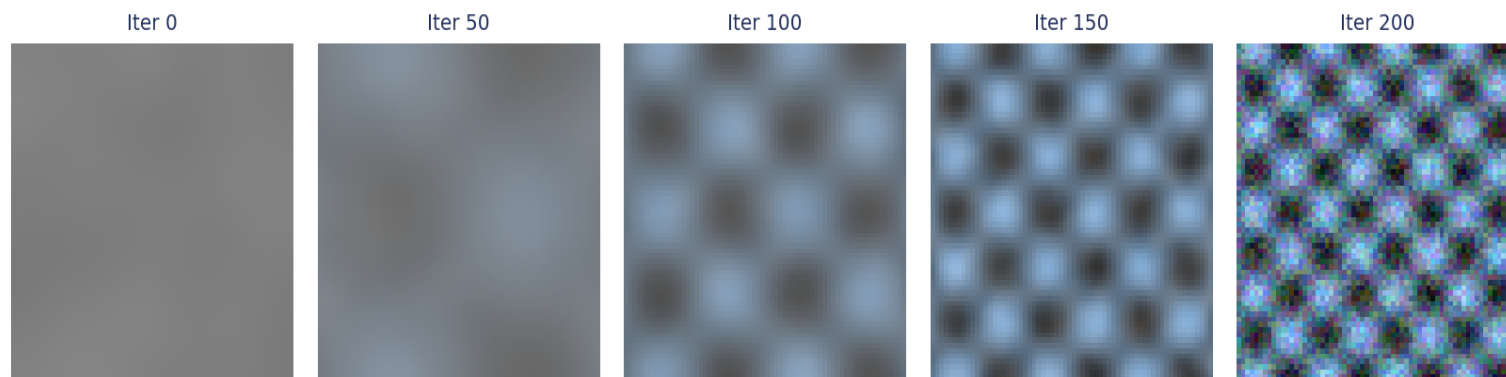
Grid position = filter index.
Pixel value = weight magnitude.
Diverging colormap for signed weights;
sequential for magnitudes only.

DESIGN RULE

Normalize per-filter before rendering.
Otherwise one large-magnitude filter
dominates and the others read as
noise — information is hidden, not lost.

Activation Maximization — Synthesis as Viz

Activation Maximization: Synthesizing the Neuron's Ideal Input



Gradient ascent on neuron activation → image that 'excites' filter k

Feature visualization asks: what is this neuron tuned for? The answer is an image we can look at.

PATTERN

Synthesis-as-visualization:
generate a prototypical input that
describes a concept. The picture IS
the explanation — no overlay needed.

PRIORS = READABILITY

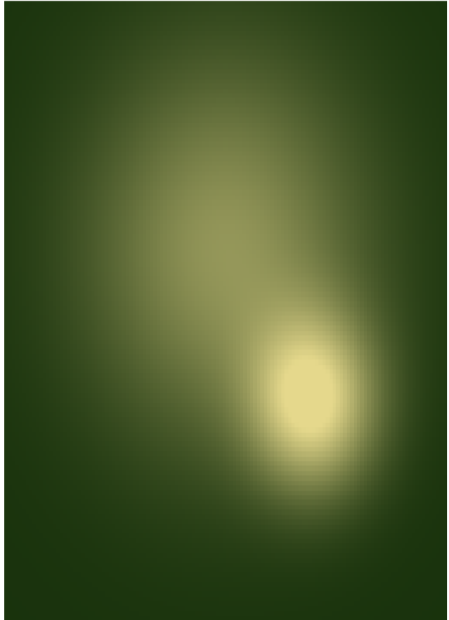
Jitter, total-variation penalty, color
decorrelation — these are VIZ READABILITY
constraints, not algorithm tricks. Without
them the output is adversarial noise.

REFERENCE

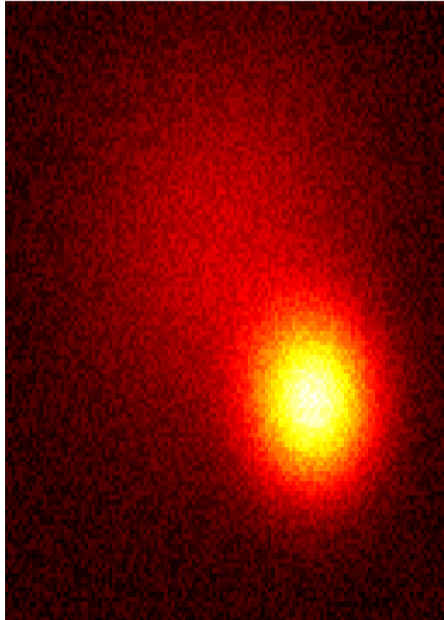
Olah, Mordvintsev, Tyka (Distill 2017)
"Feature Visualization" — canonical design
language for ML viz.

Saliency Heatmaps — Colormap & Contrast Choices

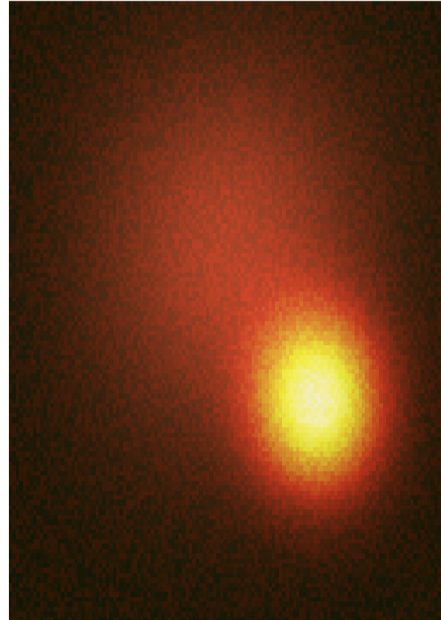
Input Image



Saliency $|\partial\hat{y}/\partial x|$



Overlay



ENCODING CHOICE

Sequential map for $|\partial\hat{y}/\partial x|$.

Diverging map for signed gradient.

Signed carries more information but readers must learn the center-point.

COLOR THEORY

Perceptually uniform maps: viridis, magma, cividis (CVD-safe).

Avoid 'jet' — it fabricates banding that readers see as data structure.

REFERENCE

Kovesi (2015), Moreland (2009).

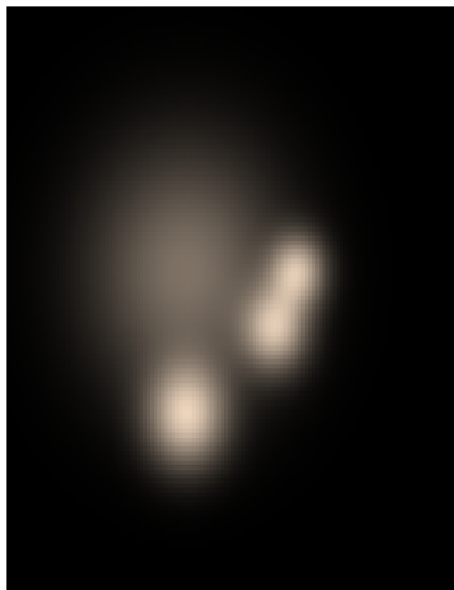
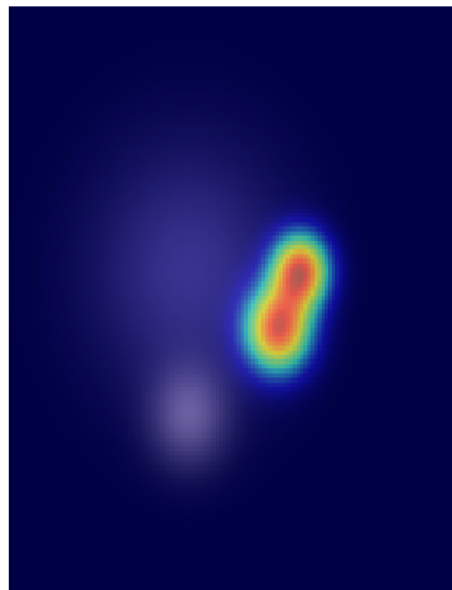
Good colormaps are not aesthetic — they are perceptual calibration.

Python

```
# matplotlib — choose the colormap deliberately
plt.imshow(saliency, cmap='viridis')      # magnitude only
plt.imshow(signed_grad, cmap='RdBu_r',    # signed
            norm=TwoSlopeNorm(vcenter=0))
```

Grad-CAM — The Overlay-Composition Idiom

Input: 'Doctor?'

Weighted Feature Maps $\sum_k \alpha_k A^k$ Grad-CAM → focuses on stethoscope

ID I O M

Heatmap + source overlay — the 'glance test.' Reader must see BOTH the explanation AND the input without either fighting the other.

B L E N D D E S I G N

$out = \alpha \cdot heatmap + (1-\alpha) \cdot source$.
Tune α (≈ 0.4) for perceived contrast.
Gamma-correct before blending or mid-tones flatten.

$$\alpha_k^c = (1/Z) \sum_i \sum_j \partial y^c / \partial A_{ij}^k \quad L_GradCAM = ReLU(\sum_k \alpha_k^c \cdot A^k)$$

the math runs in the backprop; the VIZ lives in the overlay

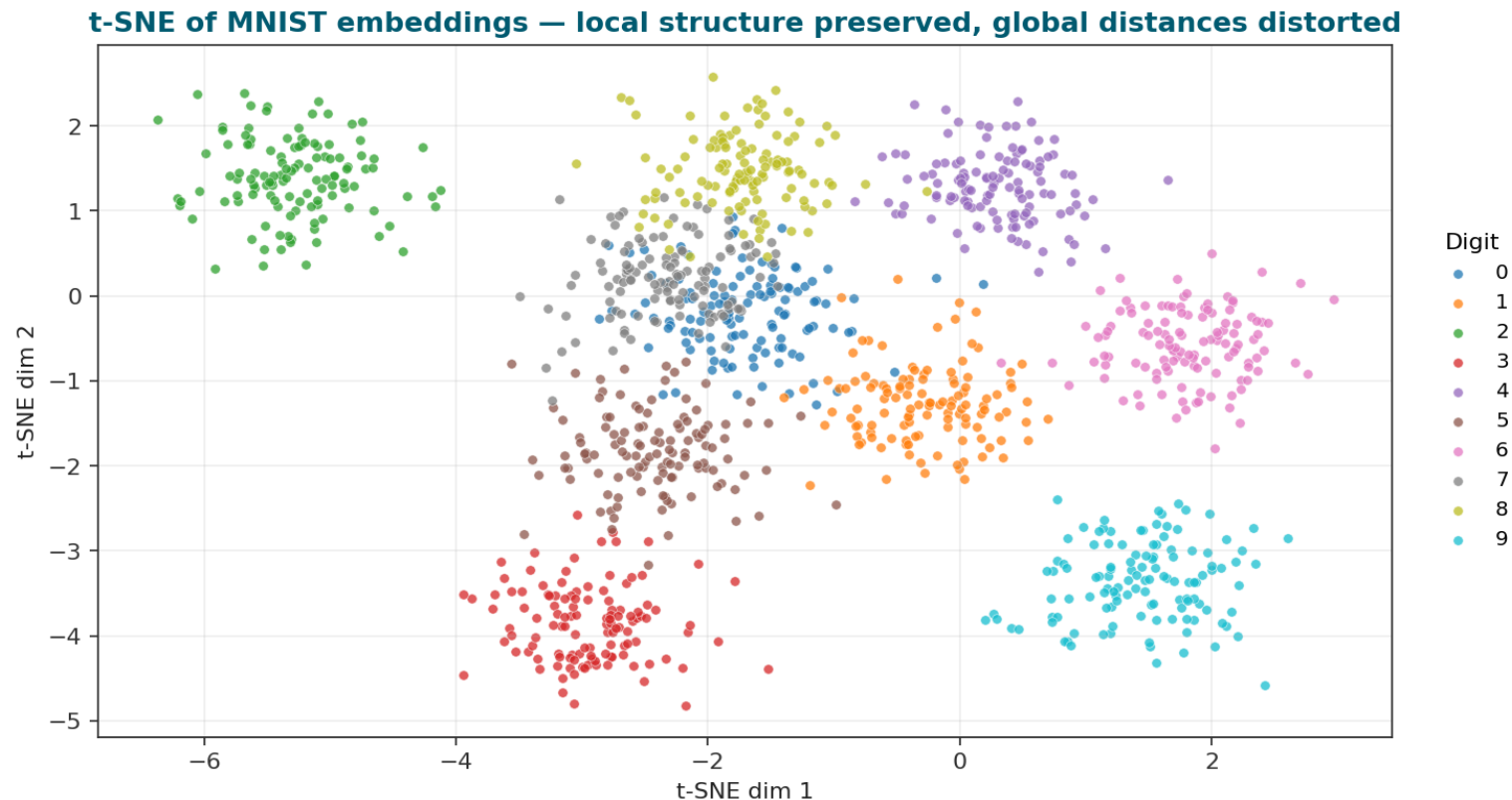
Python

```
overlay = cv2.addWeighted(heatmap, 0.4, img, 0.6, 0)
```

C A S E

Selvaraju (2017): 'doctor' classifier keyed on stethoscope, not face.
Viz made bias LEGIBLE — code alone wouldn't have surfaced it.

t-SNE — The 2D Projection Idiom



IDIOM

Scatterplot of projected embeddings.
Position is the ONLY channel —
every other visual variable (color,
size) is a metadata overlay.

DESIGN KNOBS

Perplexity (5–50), init, n_iter.
These are VIZ knobs, not stats knobs.
Try several and overlay — Wattenberg
(Distill 2016) on using t-SNE well.

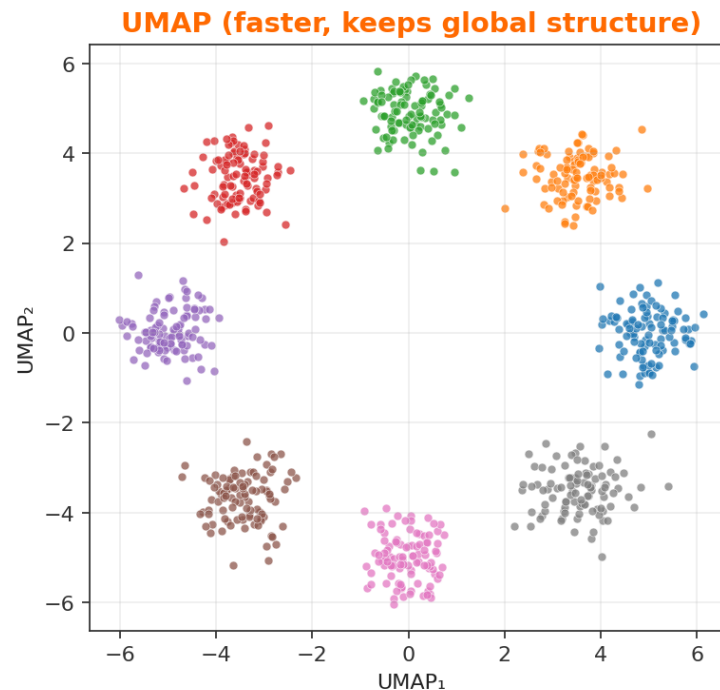
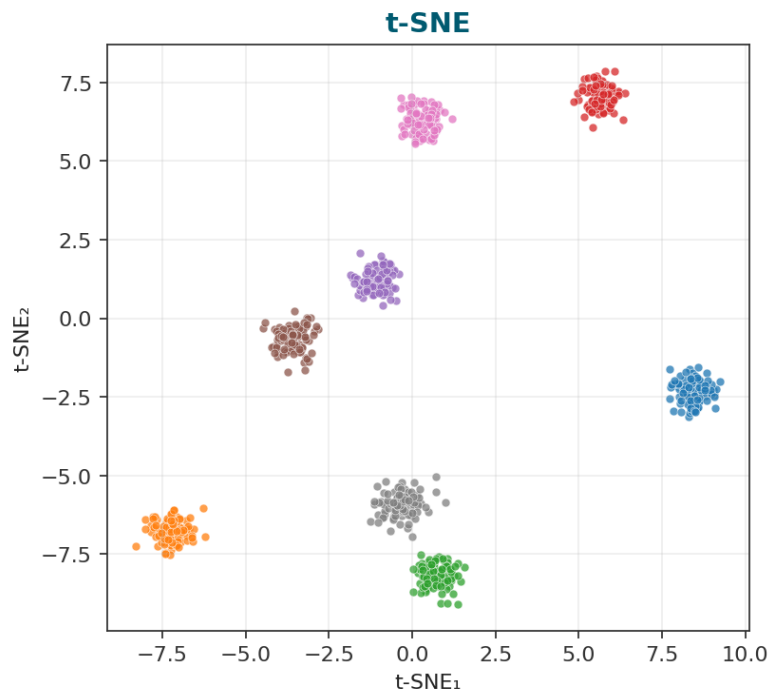
READER HAZARDS

Cluster sizes lie (density-equalized).
Inter-cluster distances lie.
Axes have no units. Rotate freely —
tell readers so they don't infer meaning.

$$KL(P \parallel Q) = \sum p_{\{ij\}} \log(p_{\{ij\}} / q_{\{ij\}})$$

the cost function drives the LAYOUT — the layout is the viz

Comparing Projections — Which Idiom, When?



LOCAL vs GLOBAL

t-SNE: local neighborhoods sharp, global distances meaningless.

UMAP: somewhat global preservation, faster at scale. Task-driven choice.

METADATA OVERLAY

Color by class = preattentive channel (Healey & Enns 1994).

Use CVD-safe palettes (ColorBrewer, Viridis categorical).

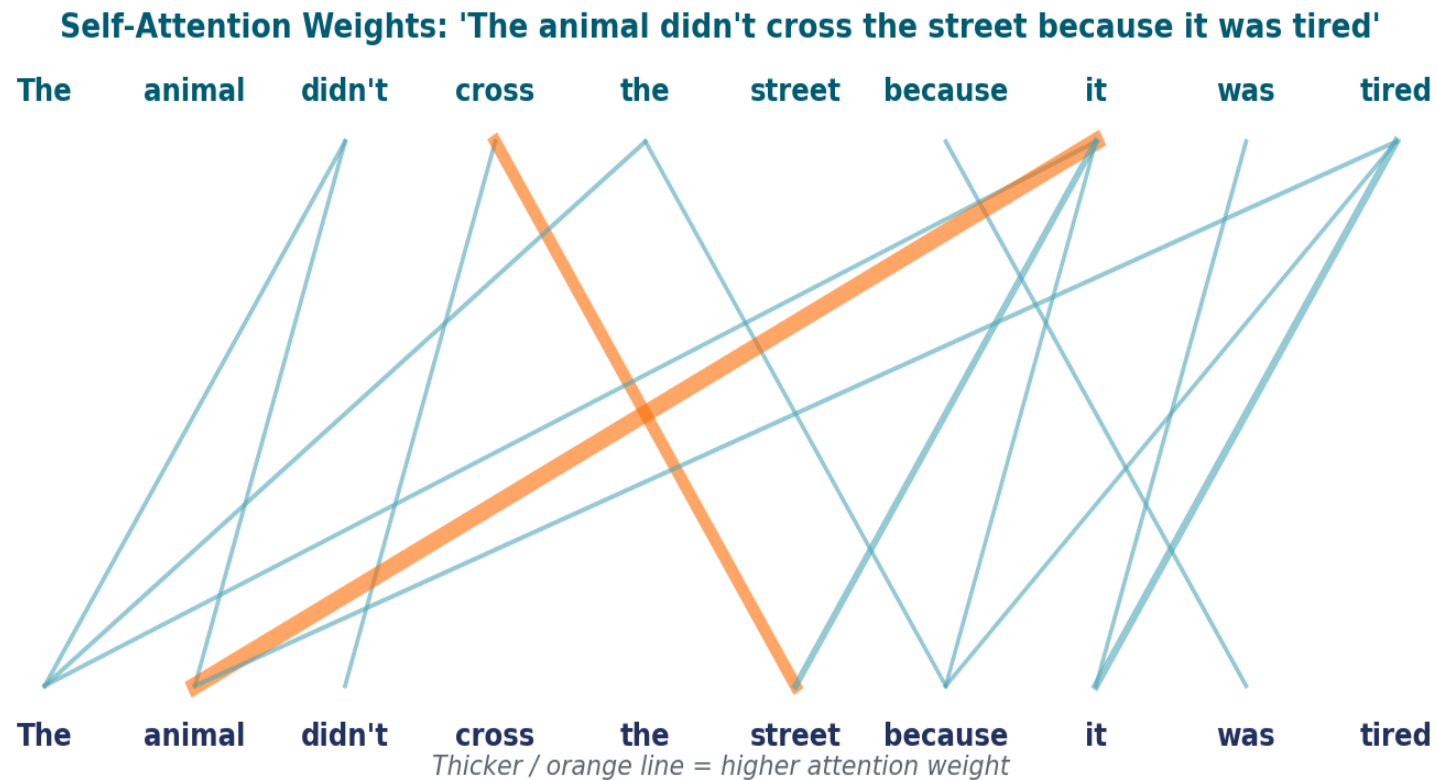
WATTENBERG'S RULE

Always render MULTIPLE projections with different hyperparameters. A single t-SNE is a hallucination.

Python

```
# Compare projections side by side – small multiples
for perp in [5, 30, 50]:
    z = TSNE(perplexity=perp).fit_transform(X)
    px.scatter(x=z[:,0], y=z[:,1], color=labels,
               color_discrete_sequence=px.colors.qualitative.Safe)
```

Attention — Matrix or Bipartite? Pick by Task



"The idiom is not the data. Choose what serves the reader's question."

TWO IDIOMS

Matrix heatmap: row=query token, col=key token, cell=weight.

Bipartite graph: arrows bundle from token to token. Same data, different task.

WHEN EACH

Matrix: dense overview, any-to-any.

Bipartite: sparse attention, top-k only.

BertViz (Vig 2019) offers both.

Let the reader switch.

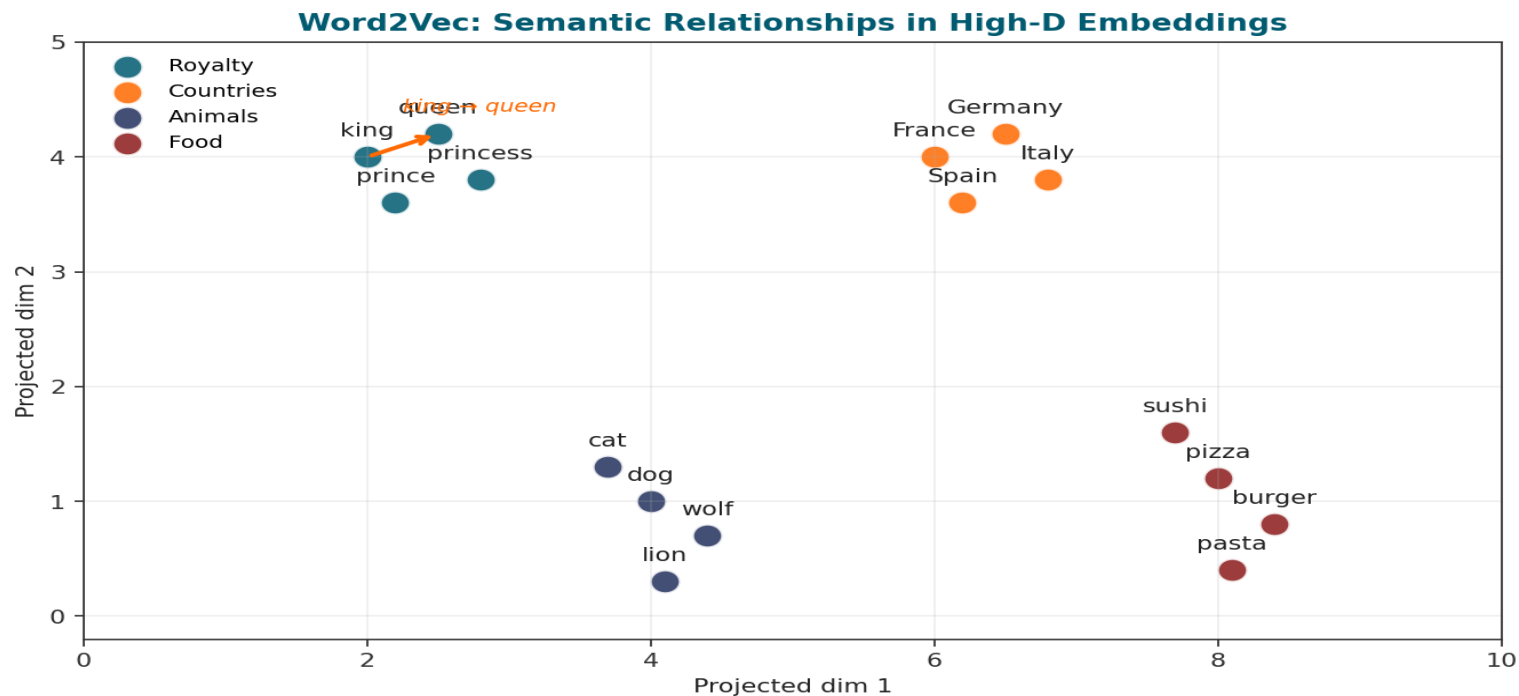
HAZARD

Beautiful does not mean faithful.

Jain & Wallace (2019): attention can be manipulated without changing output.

Viz can mislead. Label uncertainty.

Embedding Projector — Shneiderman's Mantra



Python

```
# TensorBoard projector config – viz is configuration
from tensorboard.plugins import projector
cfg = projector.ProjectorConfig()
e = cfg.embeddings.add()
e.tensor_name = 'word_embeddings'; e.metadata_path = 'meta.tsv'
projector.visualize_embeddings(writer, cfg)
```

MANTRA (1996)

Overview first,
zoom and filter,
details on demand.
Projector embodies all three.

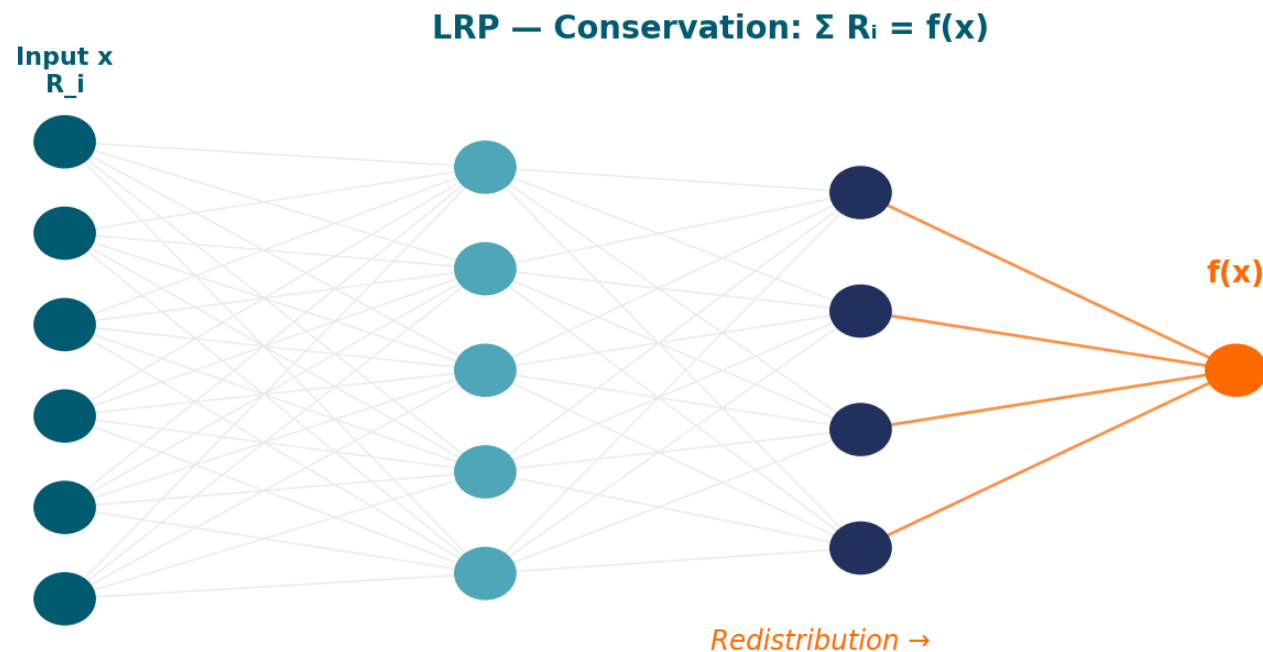
COORDINATED VIEWS

3D projection + neighbor list +
metadata panel. Roberts (2007):
linked views propagate selection.
Exploration ≠ single chart.

PROVENANCE

Saved views as bookmarks.
Reproducible exploratory viz =
the data-ink of iteration.

Layer-wise Relevance — Diverging Colormaps



$$R_j = \sum_k \left(z_{jk} / \sum_j z_{jk} \right) \cdot R_k \quad \text{conservation: } \sum R = \hat{y}$$

pixel-scale accounting — the viz colormap must honor symmetry around zero

SIGNED DATA

Relevance has + and – values.
Sequential map hides the sign.
Diverging map (red–white–blue)
makes both directions legible.

PERCEPTUAL SYMMETRY

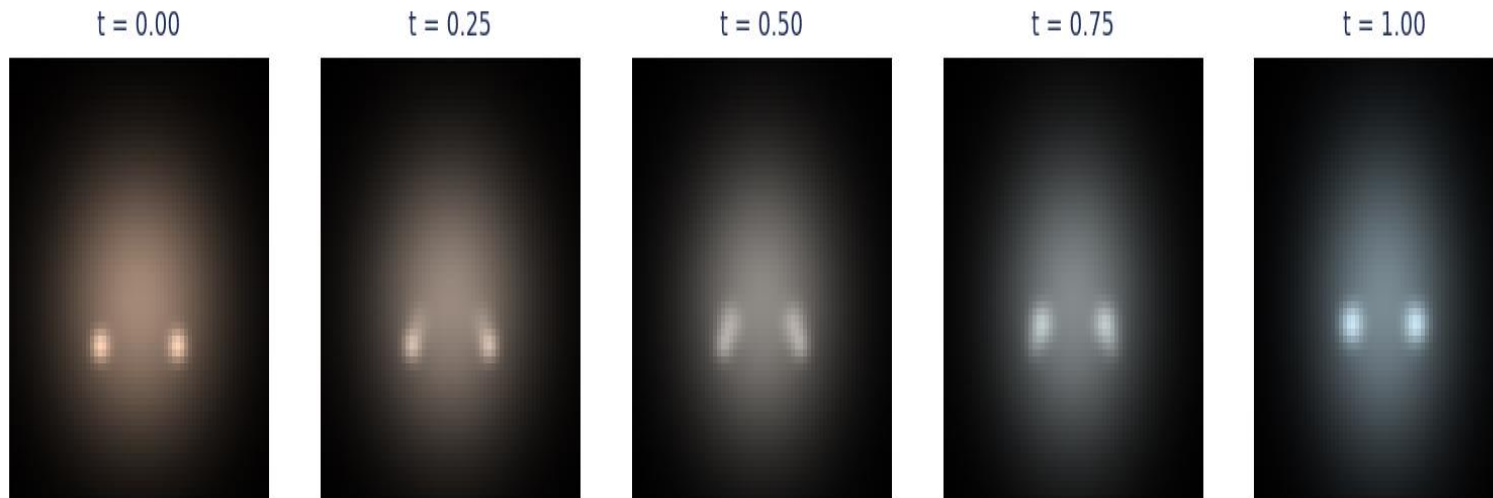
Kovesi (2015): match luminance on
both sides of the center, or readers
will over-read one direction.
Moreland (2009) — the canonical reference.

GRAD-CAM vs LRP

Grad-CAM: coarse, sequential.
LRP: pixel-level, diverging.
Different VIZ answers different questions.

Latent Interpolation — A Horizontal Small-Multiples Strip

GAN Latent Space Walk: $z_0 \rightarrow z_1$ (continuous attribute interpolation)



Python

```
# torchvision utility for strip rendering
imgs = [G(z1 + t*(z2 - z1)) for t in np.linspace(0, 1, 8)]
grid = make_grid(torch.stack(imgs), nrow=8, padding=2)
save_image(grid, 'interp_strip.png')
```

ID I O M

Strip of images at equally-spaced interpolation steps. The horizontal axis becomes a perceptual continuum for the reader to read gradient.

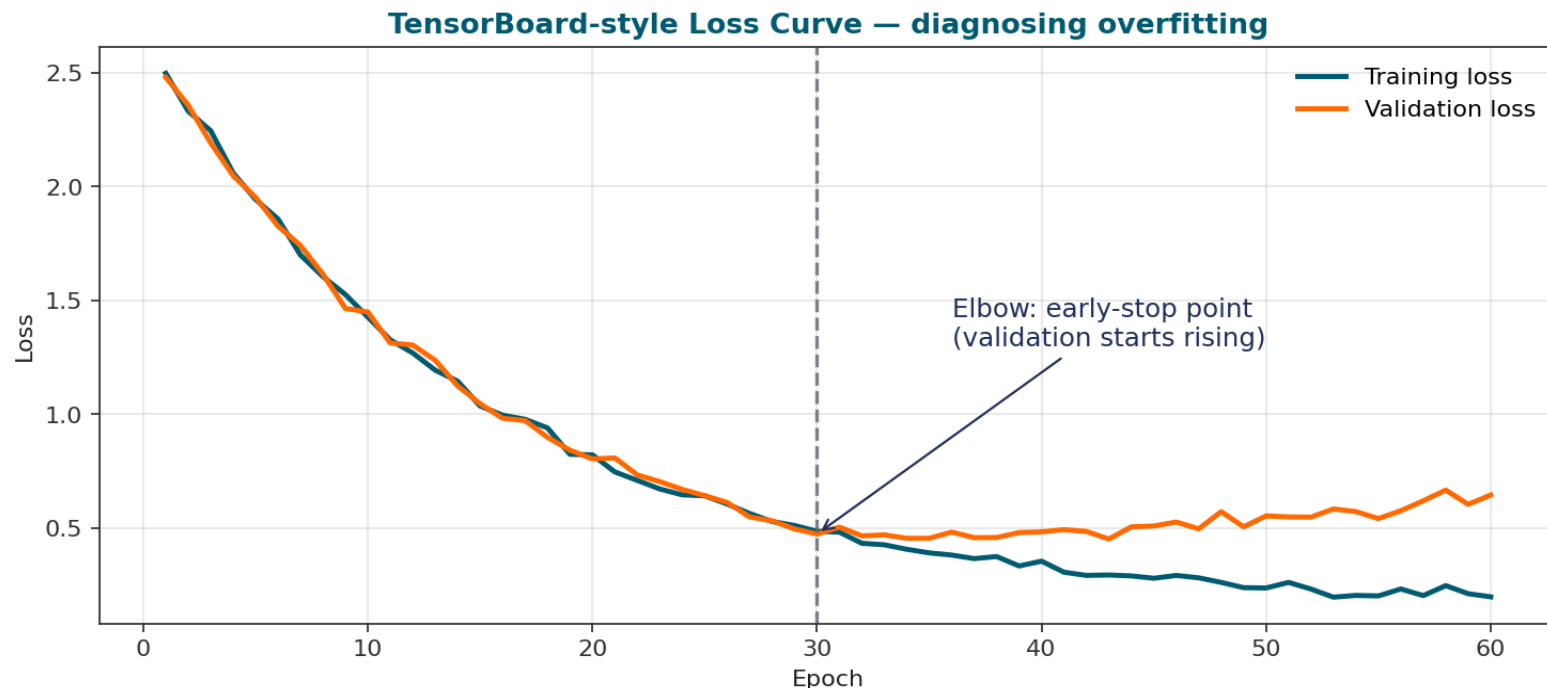
D E S I G N

Equal spacing $\rightarrow$ readers see rate of change. Uneven spacing implies INFLECTION. Both are viz choices with meaning — use deliberately.

S T Y L E G A N G R I D S

Karras (2019): 2D grids where axes = semantic directions (age, pose).
Small multiples scale to bivariate.

Training Dashboards — Coordinated Multiple Views



Python

```
# tensorboard — logs feed a CMV dashboard
writer.add_scalar('loss/train', loss, step)
writer.add_scalar('loss/val', vloss, step)
writer.add_histogram('grad/conv1', model.conv1.weight.grad, step)
# all three panels scrub together
```

CMV PRINCIPLE

Roberts (2007): multiple linked panels, where interaction propagates. Loss + grad norms + LR + histograms must share a cursor.

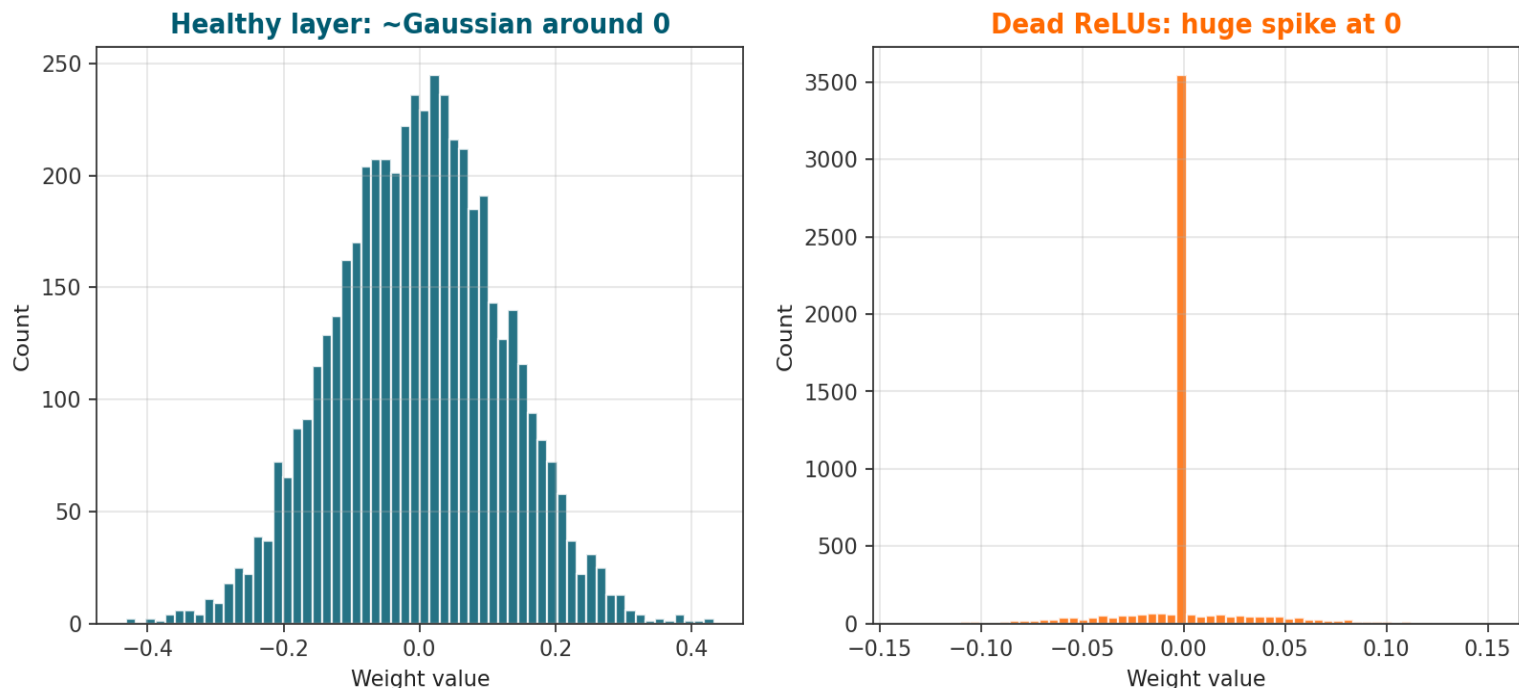
DIAGNOSTIC VIZ

Training viz is not pretty — it is DECISION SUPPORT. Every element must either help you keep training or make you stop. Otherwise it's chartjunk.

ANNOTATION LAYER

Mark the elbow. Mark the LR drop.
Tufte: data-ink + narrative ink.
Blank curves teach nothing.

Distributions — The Idiom Ladder



Python

```
# seaborn / matplotlib – the idiom is a one-liner swap
sns.histplot(w, bins=50)           # histogram
sns.violinplot(x='layer', y='w', data=df) # violin
# or joypy.joyplot(df, by='layer')      # ridgeline
```

IDIOM LADDER

Histogram → density → violin → ridgeline. Each reveals different features. The scale of comparison dictates the idiom.

SELECTION RULE

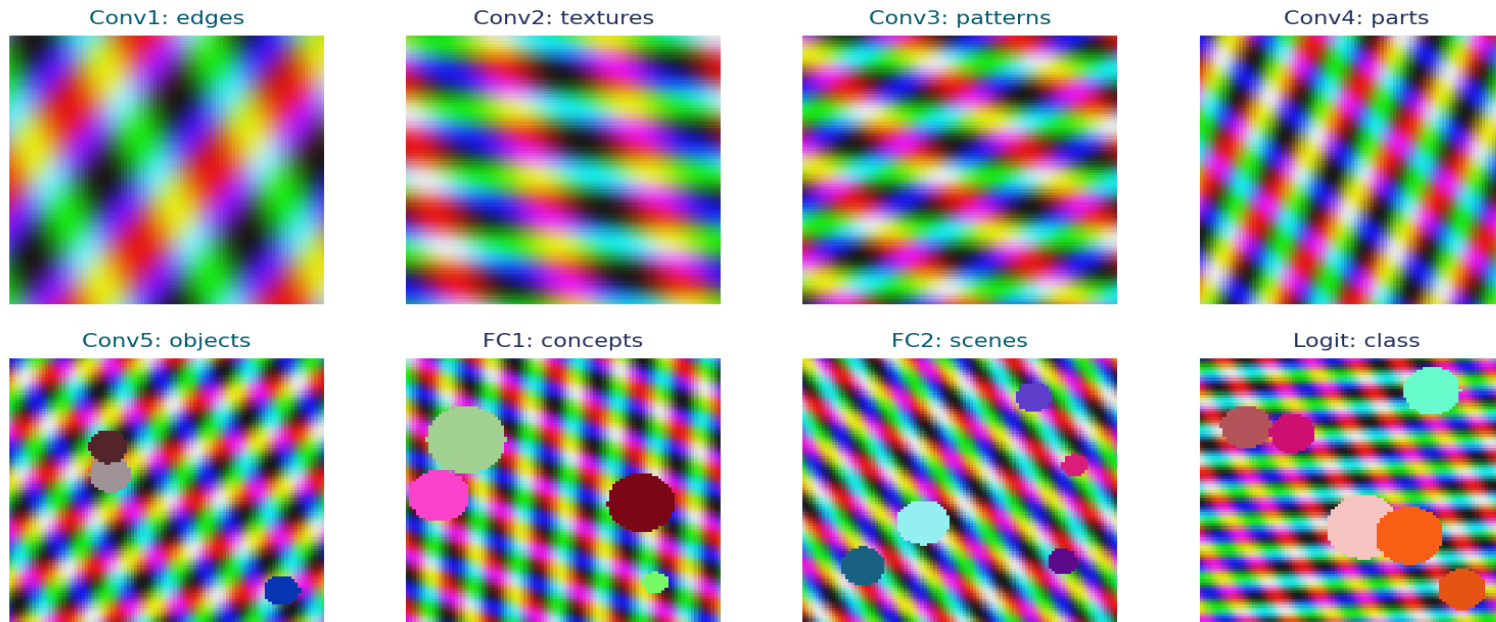
2 distributions: overlay or paired.
 3–10: small-multiples grid.
 Many: ridgeline (Wilke 2017, ggribes) — stack and stagger.

OVERPLOTING

Transparent fills, jitter, stripplots.
 Always print the N per distribution — small samples lie with smooth curves.

Feature Visualization — The Distill Design Language

Lucid / OpenAI Microscope: Feature Visualization Across Layers



JavaScript

```
// d3 sketch – hoverable feature gallery
d3.selectAll('.neuron')
  .on('mouseover', (e, d) =>
    showPanel(d.featureViz, d.topActivations))
// interactivity is the narrative
```

CONVENTION

Distill.pub pioneered interactive web essays: widgets inline with prose. Viz becomes NARRATIVE — not illustration of an argument but the argument itself.

DESIGN CHOICES

Typography: serif body, mono code.
Muted palette with single accent.
Generous whitespace. Readers should zoom IN, not scan past.

REFERENCE

Olah (2018) "Building Blocks of Interpretability" — set the aesthetic standard for ML visualization.

Visualizing the Invisible — Adversarial Perturbations

Panda (57.7% conf.)

+ 0.007 · sign($\nabla_x J(\theta, x, y)$)

Gibbon (99.3% conf.)



Python

```
# three-panel adversarial viz
fig, ax = plt.subplots(1, 3, figsize=(9, 3))
ax[0].imshow(orig); ax[0].set_title('original')
ax[1].imshow(50 * delta, cmap='RdBu_r'); ax[1].set_title('δ × 50')
ax[2].imshow(adv); ax[2].set_title('adversarial')
```

CHALLENGE

The perturbation IS invisible ($\epsilon \leq 8/255$). How do we make the attack legible on a page?
This is the core viz problem.

TECHNIQUES

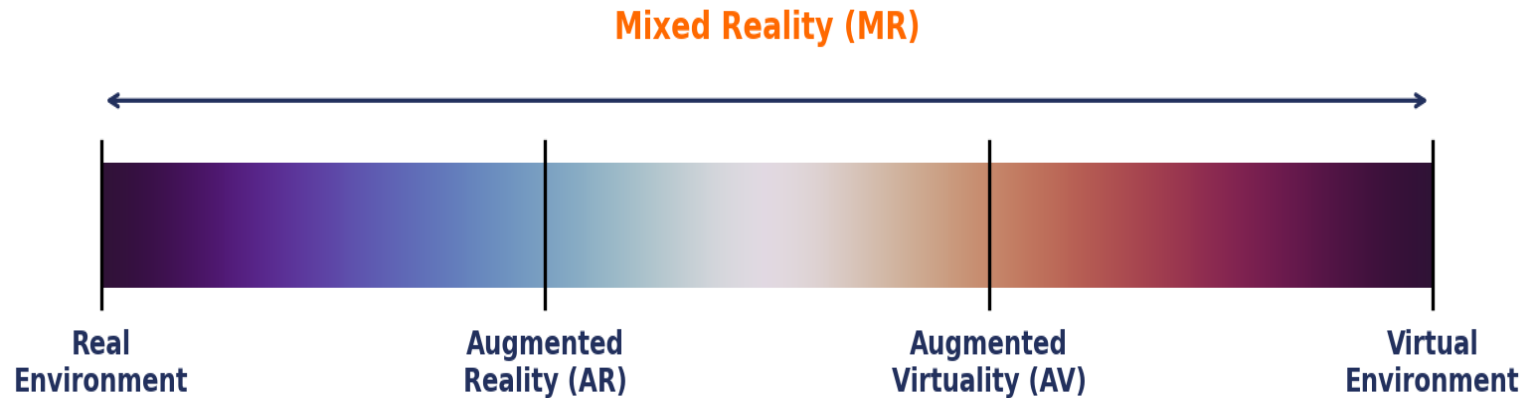
× 50 magnification of δ .
Diverging colormap of δ (signed).
Three-panel strip: original | δ | adv.
Small multiples again.

CANONICAL VIZ

Goodfellow (2014): panda → gibbon.
One of the most-reproduced figures
in DL. A teaching image, not evidence.

The Immersion Design Space

Milgram's Reality-Virtuality Continuum (1994)



"Choose the position on the continuum that serves the reader's task — not the one that dazzles them."

MILGRAM'S CONTINUUM

Real → AR → AV → VR (1994).
For viz, this is a DESIGN SPACE.
Position on the continuum is a deliberate viz choice.

TASK MATCH

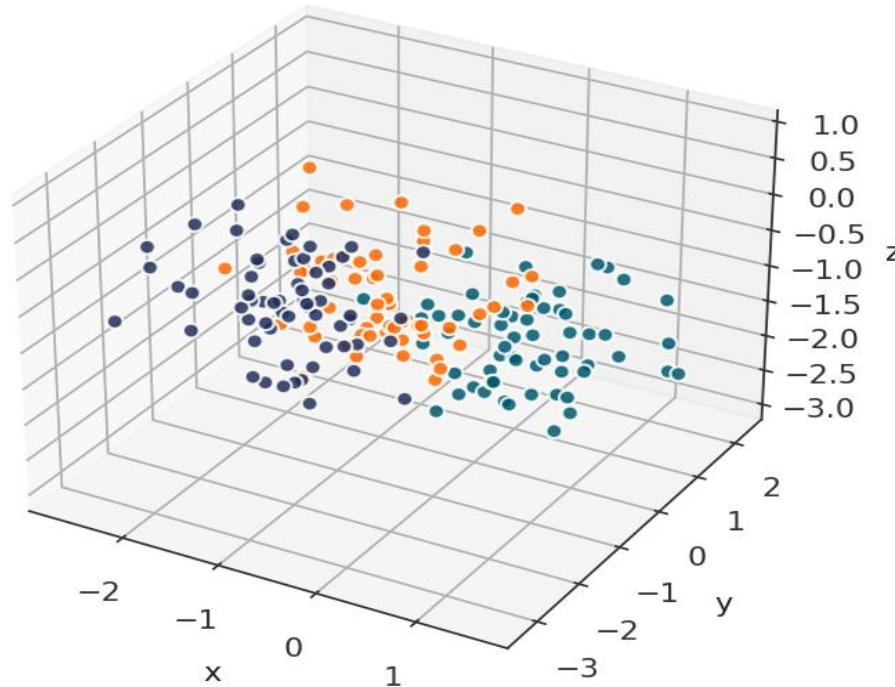
Situated analytics: AR (anchor data to its physical referent).
Spatial analytics: VR (inhabit the data). Precision reading: 2D desktop.

FAILURE MODE

"VR for VR's sake."
3D bar charts on 2D slides are almost always WRONG. Immersion must earn its keep.

When 3D Beats 2D — A Task-Driven Choice

Immersive Analytics: spatial memory anchors data to physical locations



$$effective_encoding = channel_rank \times task_match \times perceptual_ease$$

Mackinlay (1986) — encoding effectiveness is task-relative

3D WINS WHEN

Data has TRUE 3D structure:
anatomy, fluid flow, molecular
shape, architecture. The viz
axes ARE the data axes.

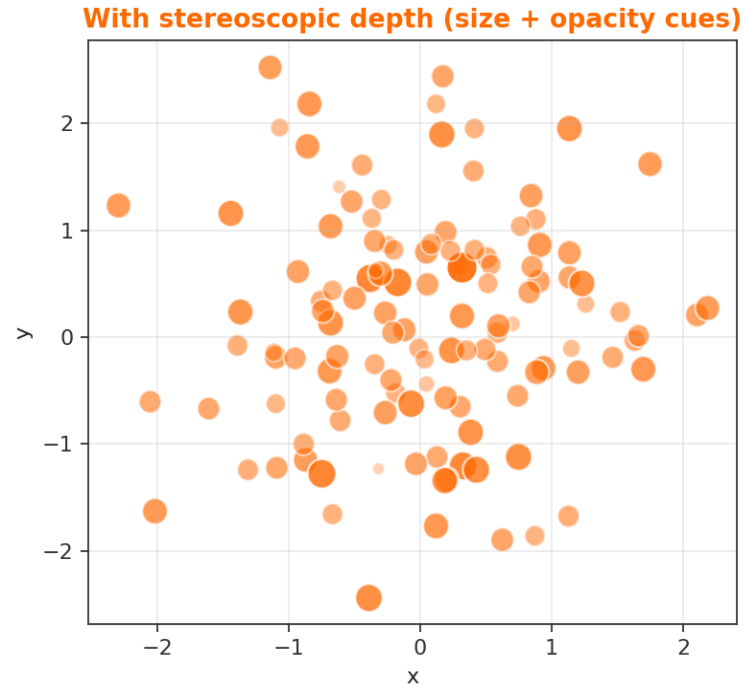
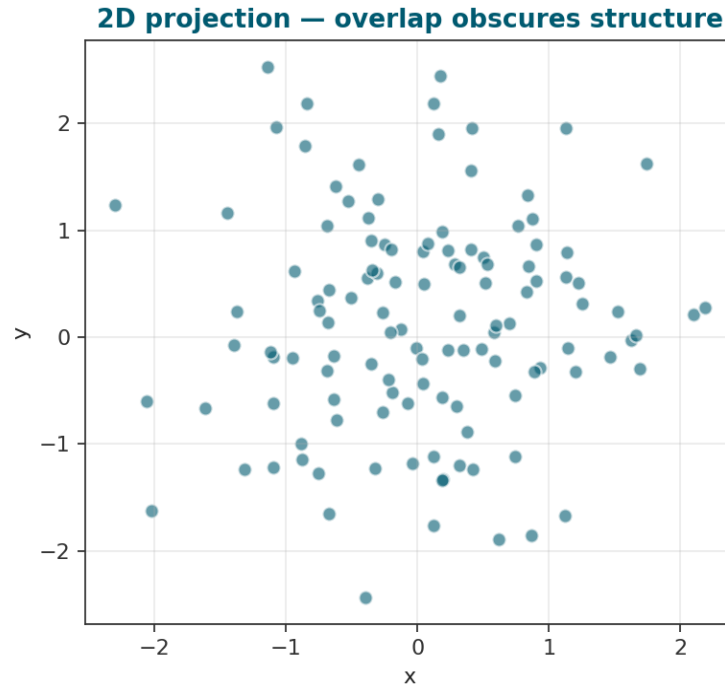
2D WINS WHEN

Comparison tasks. Precise reading.
Overview. Bertin's POSITION channel
is strongest in a plane — depth
degrades the primary encoding.

EVIDENCE

Marriott et al. (2018):
3D wins for recall in spatial tasks.
2D wins for precision. Pair them —
coordinated views across dimensions.

Bertin in 3D — New Channels, Amplified Hazards



channel_bandwidth(3D) = position > size > orientation > color > motion > depth

depth is the LEAST reliable channel — reserve it for true spatial data

NEW CHANNELS

Depth, parallax, motion, stereopsis, proprioception.
Ware (2004) extended Bertin's retinal variables for 3D.

OLD HAZARDS, WORSE

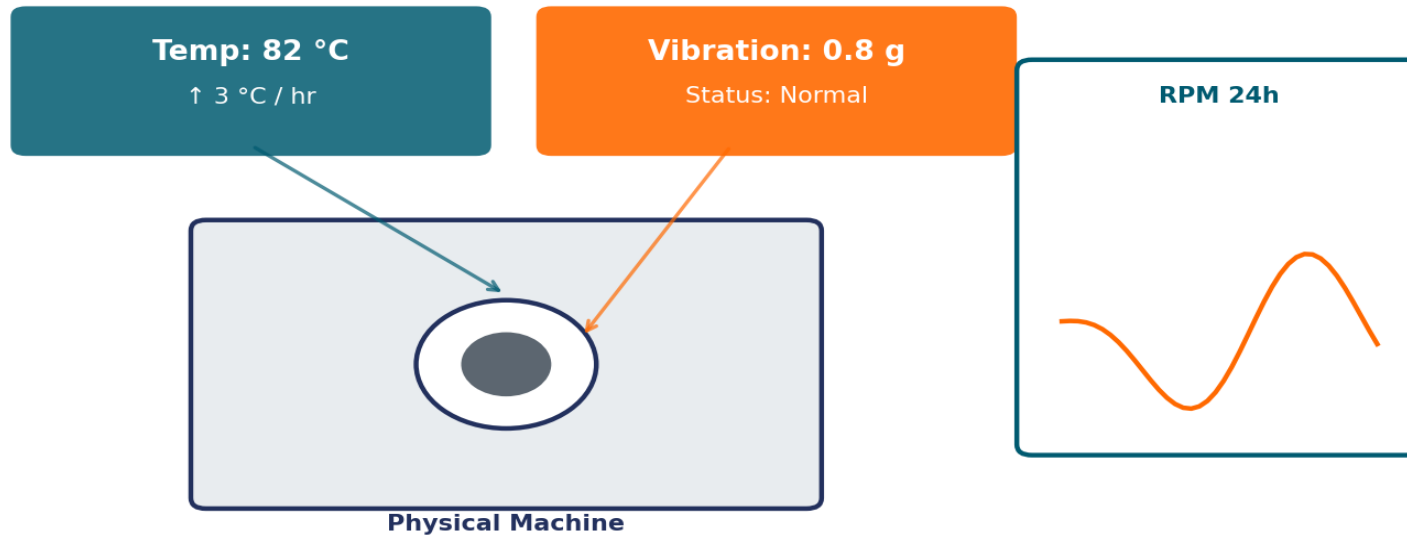
Occlusion = data loss.
Perspective distortion → area becomes unreliable. Readers misjudge distances by 20–40%.

DESIGN RULE

Position (X, Y, Z) is still king.
Don't double-encode onto depth if depth already carries data.

Situated Analytics — Anchoring Data to Objects

Situated Visualization — data anchored to the real object



"The physical world is the most information-rich background any visualization will ever have. Don't fight it — design with it."

DEFINITION

Elmqvist & Irani (2013):
situated viz — data rendered IN
the physical context it describes.
The world is part of the chart.

DESIGN CHALLENGE

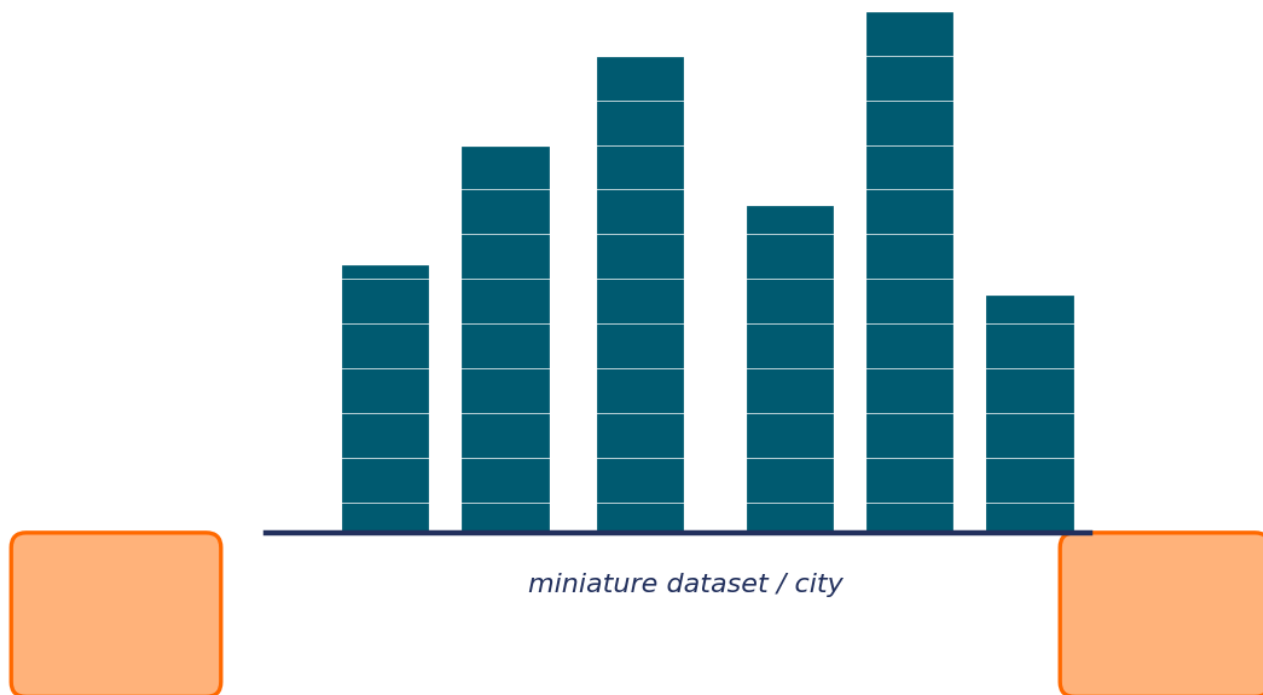
AR viz CANNOT occlude the objects
it describes. Negative space is a
primary channel. Labels float, not block.

CASES

Museum object labels (Vision Pro).
Warehouse picker overlays.
Surgical navigation markers.

Exocentric Views — The World-in-Miniature Idiom

Exocentric 'God View' — miniature world in the user's hands



Shneiderman's mantra, in VR: the WIM is the overview.

WIM IDIOM

Stoakley et al. (1995):
a small replica of the scene held
in hand. Overview + navigation.
Map metaphor, made literal.

ENCODING IN 3D

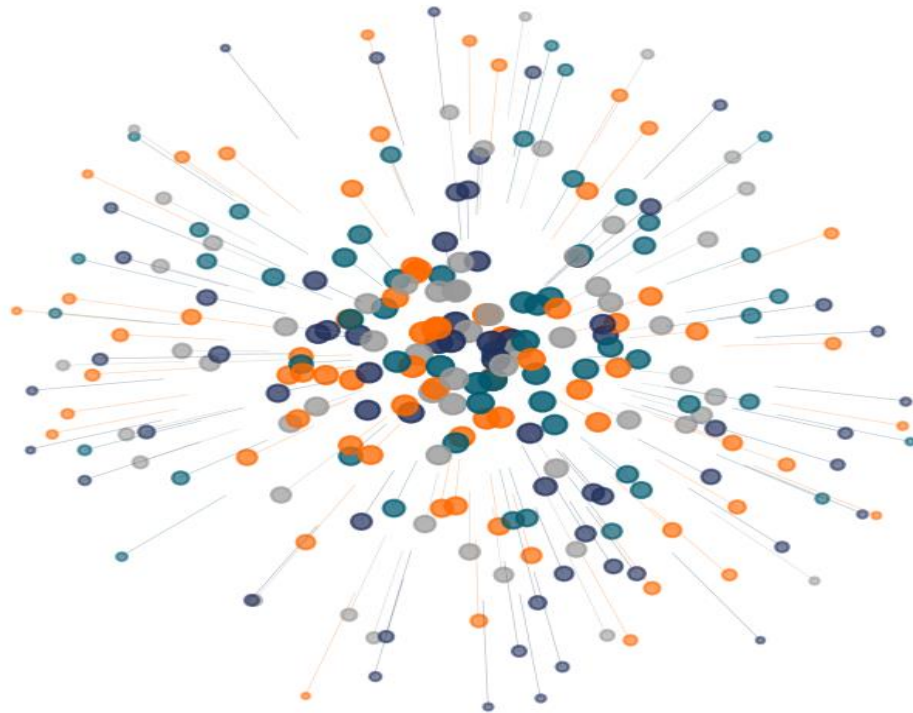
Top-down projection remains the
most efficient for comparison tasks
(position, area). 3D overviews often
collapse gracefully to 2D.

TRADE-OFF

Detail loss. The WIM is an index,
not a document. Pair it with an
egocentric detail view — coordinated
views across scales.

Egocentric Views — Detail on Demand in Space

Egocentric 'Inside-out' View — flying through a data galaxy



Immersion without structure is just being lost in data.

METAPHOR

Walk inside the data. Your body becomes the camera. Proprioception carries spatial memory the mouse cannot reach.

DESIGN HAZARD

Spatial disorientation.
Breadcrumbs, anchors, a compass — these are VIZ CUES about navigation, NOT about the data. Distinguish them.

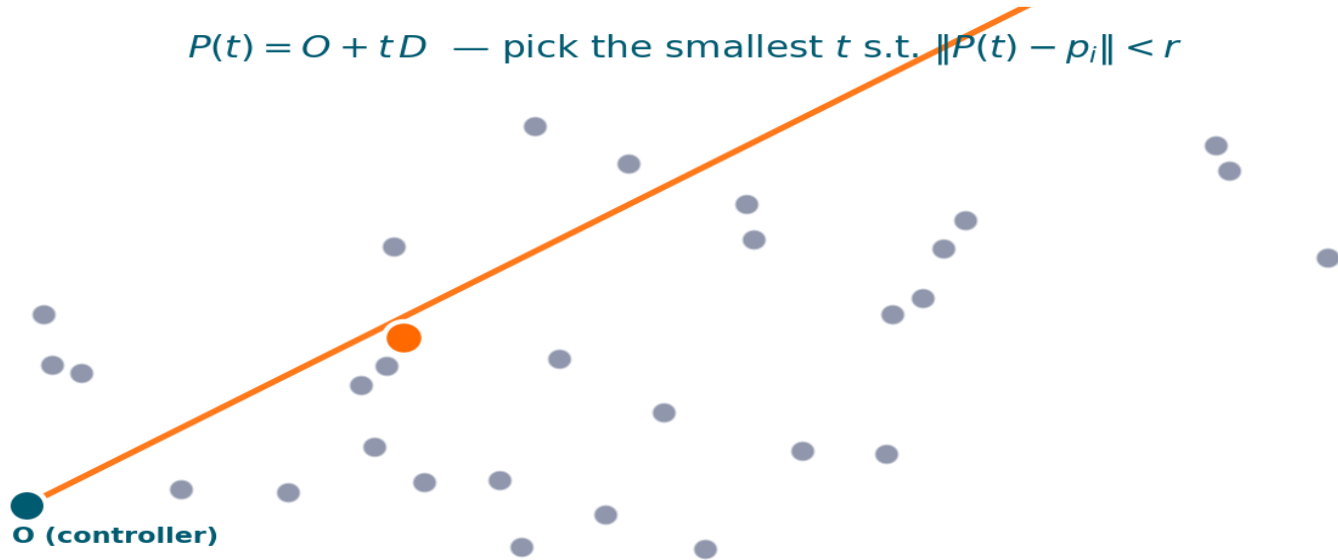
SHNEIDERMAN IN 3D

The egocentric view IS "details on demand."
Use the WIM for overview.
The mantra survives the dimension shift.

Selection in 3D Viz — Fitts + Feedback

Ray-casting Selection in VR

$P(t) = O + tD$ — pick the smallest t s.t. $\|P(t) - p_i\| < r$



FITTS' S LAW

$MT = a + b \cdot \log_2(2D/W)$.

Effective width shrinks with distance — selection at range is harder than on a desk.

VIZ FEEDBACK

The target must PRE-HIGHLIGHT under the ray. Visual feedback IS the affordance. Without it users aim blind.

ALTERNATIVES

Bubble cursor (Grossman 2005).

Precision pointer.

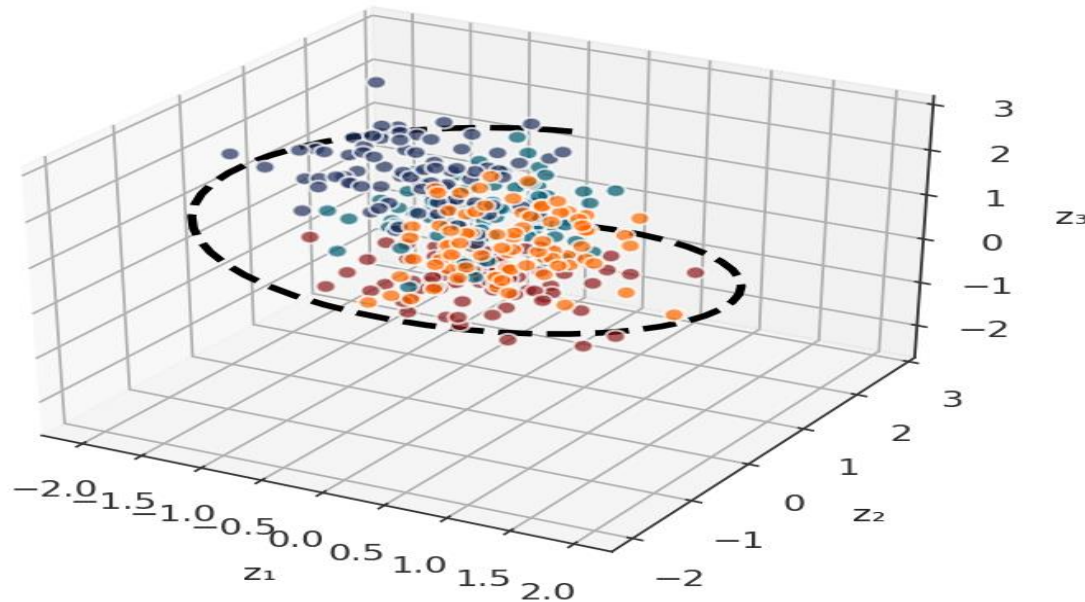
Smart snap (Vanacken 2007).

$$ID = \log_2(2D / W) \quad MT = a + b \cdot ID$$

in 3D viz, the effective W depends on visual feedback — design it in

Immersive Scatterplots — Helps & Hazards

Walking through a 3D latent manifold of image embeddings



"3D for exploration, 2D for measurement. They complement — they don't replace."

PROMISE

Walk inside a t-SNE or UMAP.
Three axes instead of two.
Occluded points reveal on parallax.
Spatial memory sticks.

HAZARD

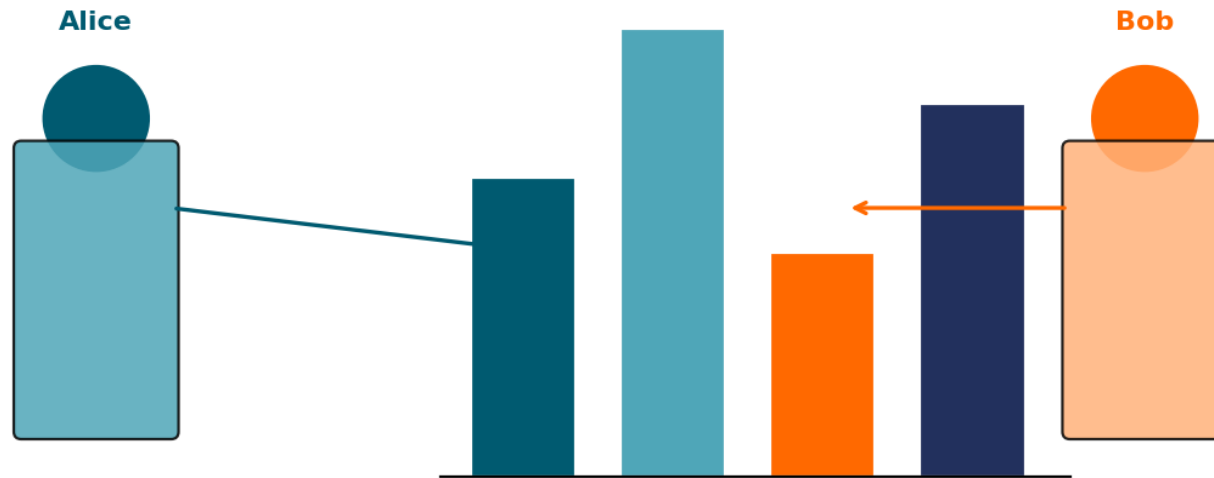
Precision reading is WORSE than 2D.
For comparison tasks, 2D wins
(Kraus 2020). Immersion $\neq$ insight.

DESIGN PATTERN

Pair immersive + 2D. Coordinated views. Best of both dimensions — reader picks the question.

Shared Viz — Coordinated Views Across Users

Collaborative VR — shared spatial reference frame for social analysis



When two people navigate the same chart, the chart is a meeting room.

NEW AXIS

CMV (Roberts 2007) extends to USERS, not just panels. Shared cursors, shared selections, shared annotations. Viz becomes social infrastructure.

AWARENESS CUES

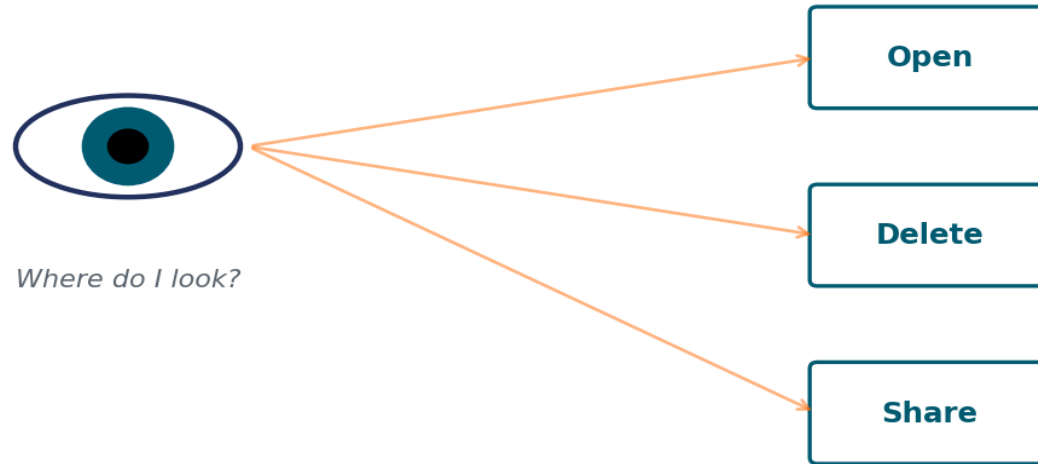
Where is my collaborator looking?
Gaze ray, avatar pointer, ghost hand.
These are VIZ elements ABOUT people, layered on viz elements about data.

PLATFORMS

Mozilla Hubs (open, WebXR).
Arthur.digital (enterprise).
ImAxes (Cordeil 2017) — data-viz specific research prototype.

Interaction Rule — Every Selection Needs Confirmation

The Midas Touch Problem



Midas Touch: if looking = clicking, every glance fires an action.

"An interface that does what the user looked at is not responsive — it is rude."

MIDAS TOUCH

Jacob (1991): if LOOKING selects, every glance becomes a click.
Implicit input without confirmation breaks the interface.

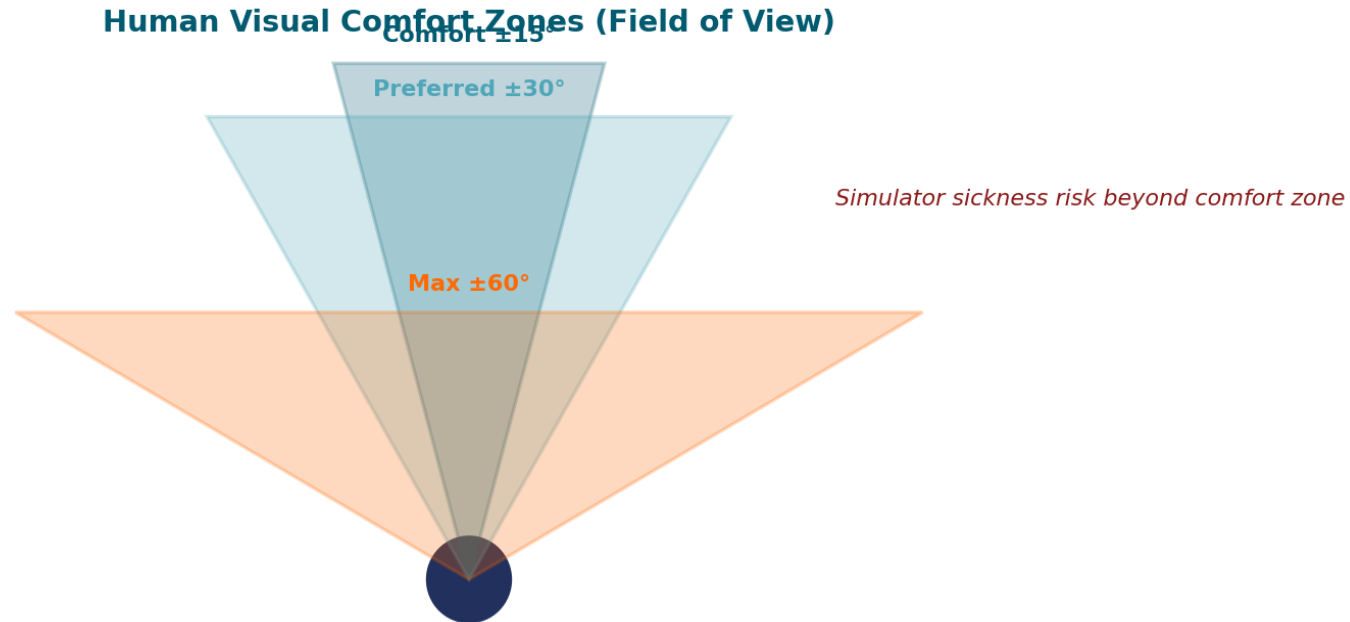
VIZ LESSON

Dashboard tooltips on scroll.
Hover-to-filter charts.
Gaze menus in VR.
SAME problem, same design answer.

ANSWER

Dwell time. Preview state.
Explicit confirmation gesture.
Undo within one step.

Design Constraints for Long-Session Viz



$$\text{min_font_height} = 2 \cdot \text{distance} \cdot \tan(0.3^\circ)$$

viewing-distance × acuity — the primary readability constraint for VR text

SESSION REALITY

Analytics sessions last 1+ hour.
Cybersickness is a VIZ FAILURE.
Design the chair before
you design the chart.

DESIGN RULES

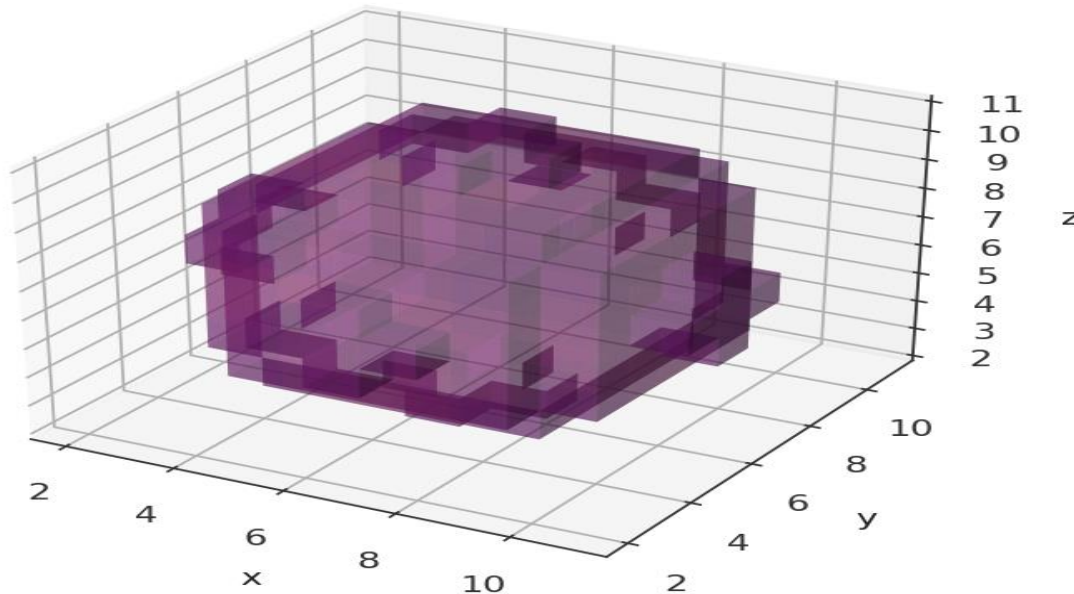
Stable horizon.
Teleport locomotion.
Seated UX option.
High-contrast text for 25-ppd displays.

READABILITY BOUND

Font $\geq 0.6^\circ$ visual angle.
At 1.5 m that's ≈ 1.6 cm high.
Smaller = subpixel mud.

Volumetric Viz — The Transfer-Function Design Problem

Volumetric (Voxel) Rendering — 3D medical CT / smoke



Python

```
# VTK transfer-function sketch
tf = vtk.vtkColorTransferFunction()
tf.AddRGBPoint( 0, 0.00, 0.00, 0.00) # air — transparent
tf.AddRGBPoint(500, 1.00, 0.78, 0.65) # soft tissue
tf.AddRGBPoint(1200,1.00, 1.00, 0.95) # bone
```

TRANSFER FUNCTION

Scalar value → (color, opacity).
THE primary viz design decision
for 3D volumes. Every rendered
frame depends on this mapping.

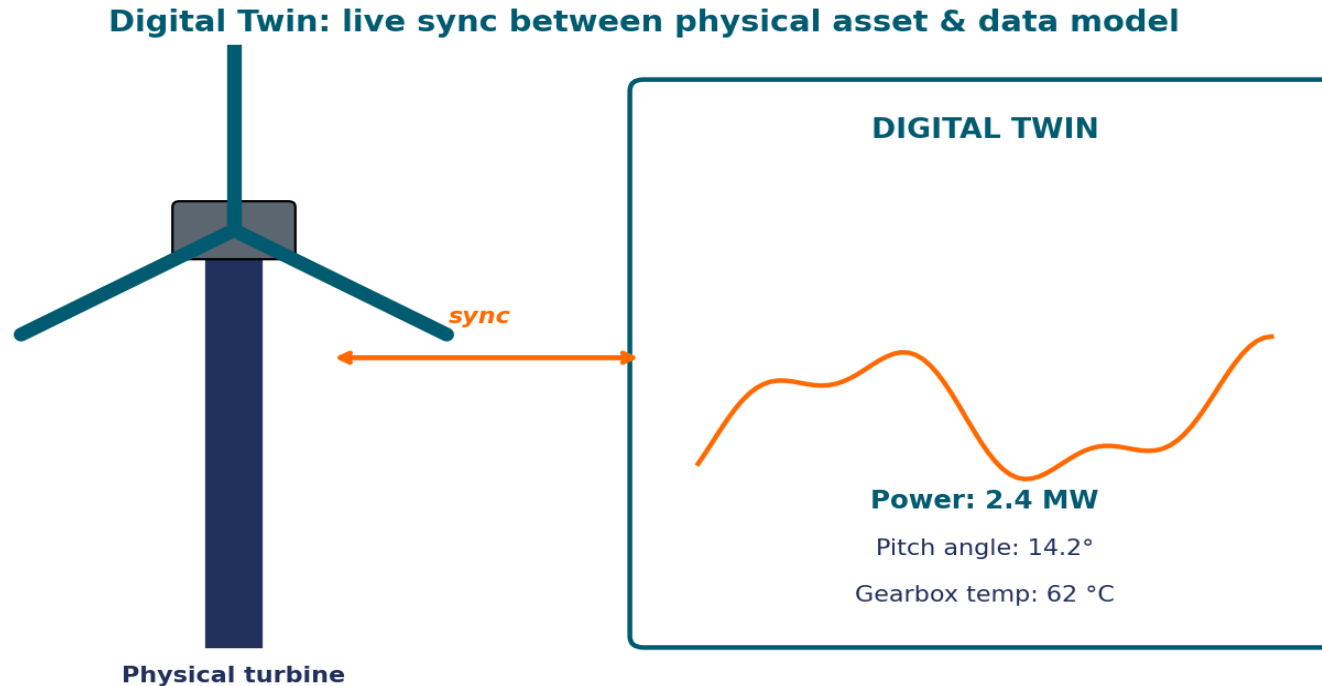
2D TF (Kindur 2003)

Map (intensity × gradient) → color.
Boundaries pop out where 1D TFs
smear. Two input axes; perceptually
richer output.

TOOLS

ParaView, 3D Slicer, Exposure
Render, Volview. Every tool has a
TF editor — it's the main viz UI.

Digital Twins — Live Viz as a Dashboard



A digital twin is not a 3D model — it is a visualization whose data updates live.

TUFTE, UPDATED

Data-ink that UPDATES.
Stale viz is worse than no viz — it confidently lies. Every element must have a freshness indicator.

TEMPORAL ENCODING

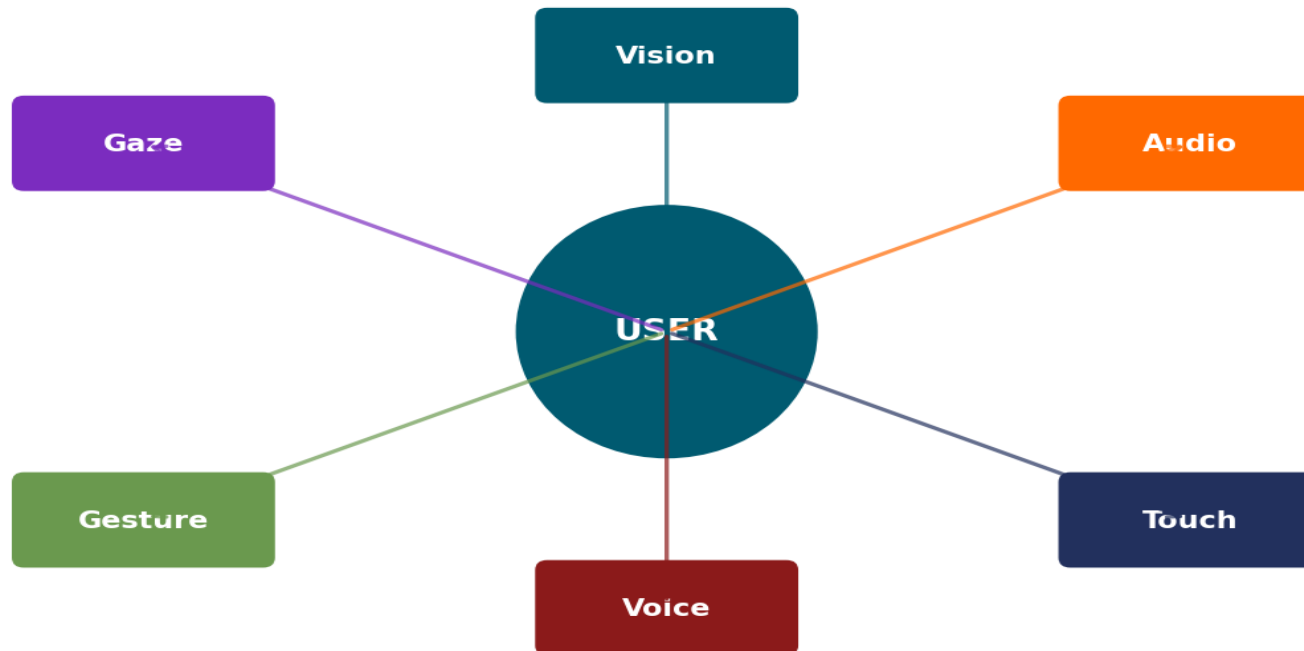
Trails, fading, tick pulses.
Time-since-update is a primary channel, not decoration.
Readers need it to trust the view.

CASES

NYC Citizen Dashboard.
Siemens factory twins.
Raz's simulation paradigms (SDS/DES/ABS/MCS) power the engine behind.

Beyond the Pixel — Channels for Encoding Data

Multimodal Interaction — parallel channels for input & output



"The chart is no longer only the pixels."

BERTIN, EXTENDED

Vision is one channel.
Audition, haptic, proprioception —
also encoding channels. Multimodal
viz is Bertin's list, grown.

CHANNEL RANKING

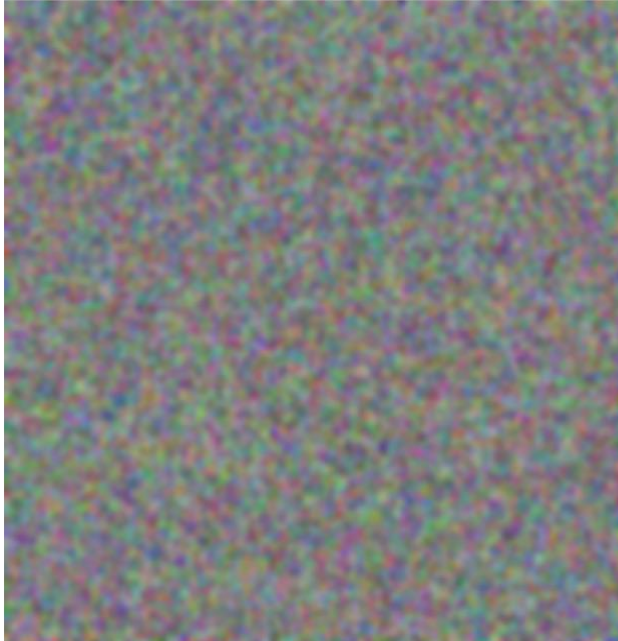
Vision: highest bandwidth.
Audio: strong for time-series, alerts.
Haptic: pre-attentive, interruption-safe.
Don't ask every channel to do everything.

DESIGN PRINCIPLE

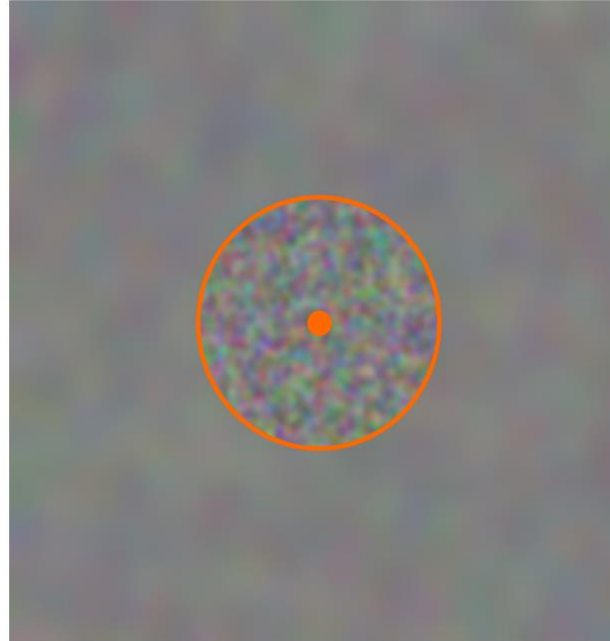
Oviatt & Cohen (2015): put each
datum on the channel best suited
for it. Channel = encoding resource,
not an accessibility afterthought.

Foveation — Progressive Disclosure via Gaze

Full-resolution render (GPU-heavy)



Foveated render — sharp only where the eye looks



JavaScript

```
// WebXR + gaze-contingent detail
xrSession.requestReferenceSpace('viewer').then(ref => {
  xrSession.requestAnimationFrame( (_, frame) => {
    const gaze = frame.getViewerPose(ref);
    scene.setLOD(gazeToScenePoint(gaze));
  })
})
```

VIZ PATTERN

Shneiderman's "details on demand" —
but the demand is GAZE, not click.
Eye position becomes the cursor.
Reveal detail without a gesture.

DESIGN

LOD sphere around fixation.
Outside: summary aggregates.
Inside: full resolution detail.
Transition must be imperceptible.

HARDWARE

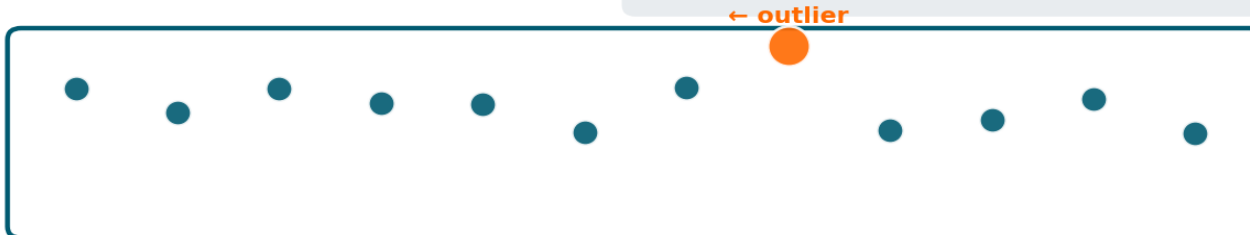
Vision Pro, Quest Pro, PSVR2.
Consumer eye-tracking is here.
Foveated viz is a shipping pattern.

NL + Chart — The Mixed-Initiative Viz Interface

Natural-Language Queries: talking to your data

“Show me the outlier in the Q3 report”

Parsed: `filter(quarter=Q3) · detect(outlier)`



The chart generated by a prompt must show WHO generated it and FROM WHAT DATA. That is a viz decision.

MIXED INITIATIVE

Horvitz (1999): user and system co-construct the result. Neither leads alone. Viz is the shared artifact they build together.

AMBIGUITY DESIGN

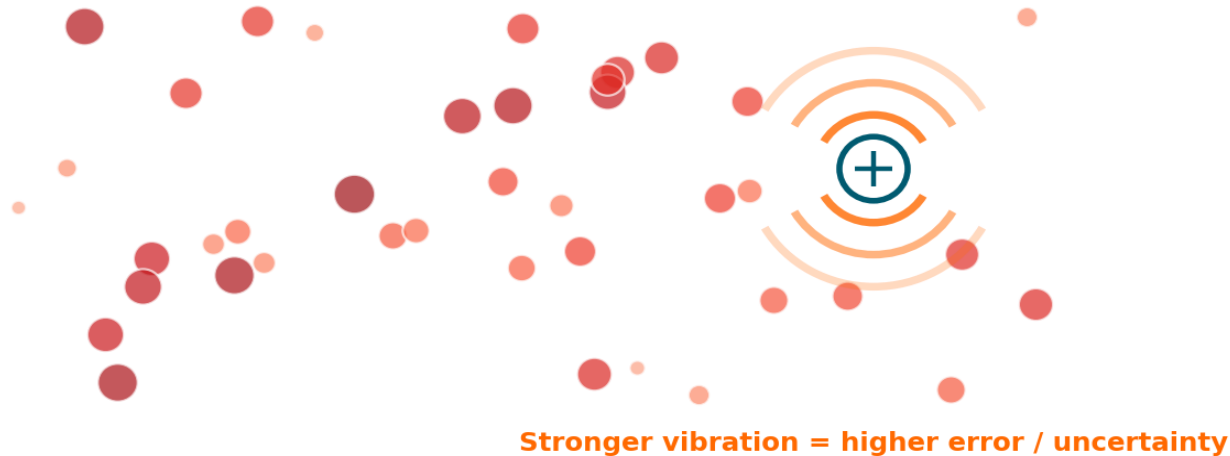
Setlur et al. (Eviza 2016): show alternate interpretations as chips. Let the user pick. Don't pretend NL has one meaning.

LLM ERA

Claude/Copilot generate charts. Provenance indicators are a viz design challenge — show the data source.

Touch as an Encoding Channel

Haptic Feedback — feeling the data



JavaScript

```
// WebXR Gamepad API — intensity encodes |error|
function onHover(dataPoint) {
  const err = Math.abs(dataPoint.residual);
  const amp = Math.min(1, err / MAX_ERR); // 0-1
  controller.hapticActuators[0].pulse(amp, 120);
}
```

CHANNEL PROPERTIES

Haptic bandwidth $\approx$ 50 bit/s.
Amplitude, frequency, texture,
spatial location — all encoding
dimensions, like color and shape.

ENCODING RULES

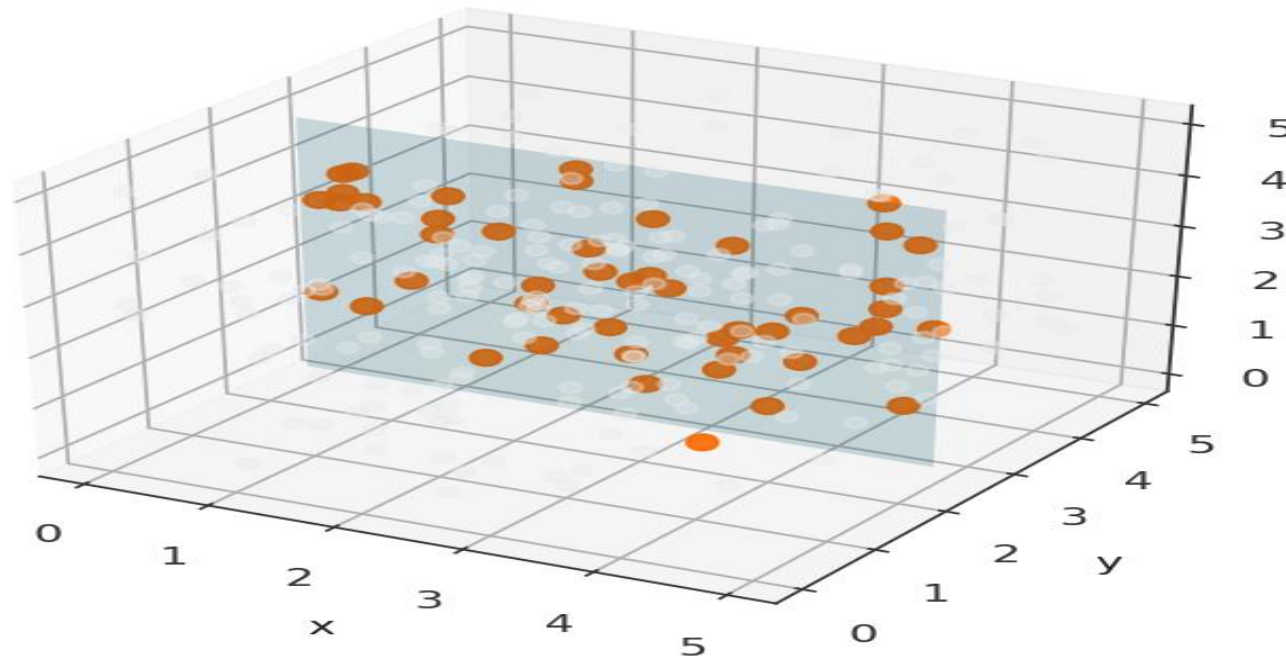
Amplitude $\propto$ ordinal magnitude.
Frequency $\propto$ category distinction.
Spatial position $\propto$ location on chart.
Treat haptic like you'd treat color.

ACCESSIBILITY

Haptic + sonification = screen-reader
viz for blind/low-vision readers.
Not an add-on — a full alternate idiom.

Direct Manipulation — Gesture Affordances in Viz

Hand-gesture slicing through a 3D data cube



"Gestures that look natural in a demo often fail in the 8th hour."

DIRECT MANIPULATION

Hutchins, Hollan, Norman (1985):
continuous representation, rapid
incremental action, reversibility.
The chart IS the interface.

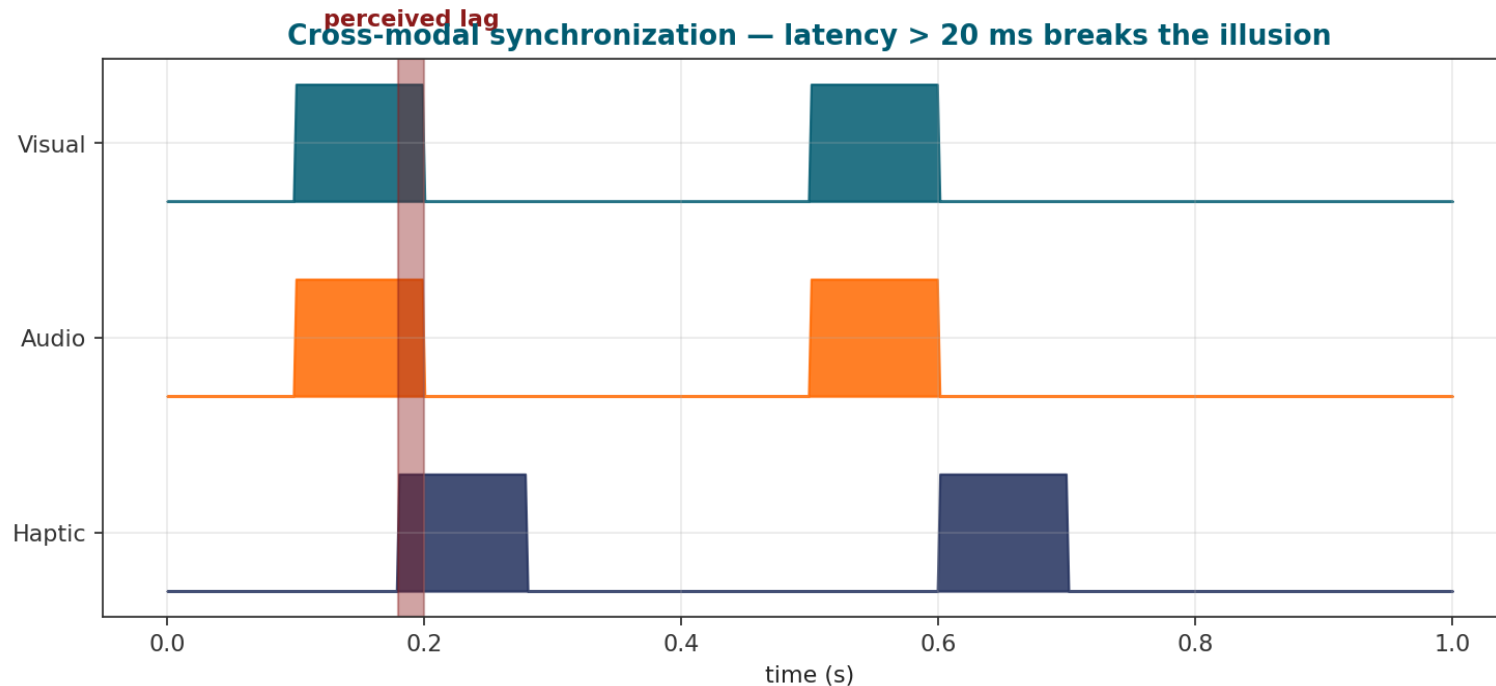
TOUCH vs MID-AIR

Touch has a surface (haptic ground truth). Mid-air has none — error doubles. Show a virtual surface when gesturing in free space.

VOCABULARY DESIGN

Pinch-zoom is learned.
"Export" is invented.
Every gesture costs something to teach — pick a small set.

Cross-Modal Sync — Temporal Binding as Viz Design



$$\Delta t_{\text{perceived}} = |t_{\text{visual}} - t_{\text{audio}} - t_{\text{haptic}}| < 20 \text{ ms}$$

channel binding is a viz design constraint, not an engineering afterthought

PERCEPTUAL THRESHOLDS

<20ms: imperceptible.
 20–100ms: tolerable.
 >100ms: broken.
 Design to the 20-ms bound.

M c G U R K E F F E C T

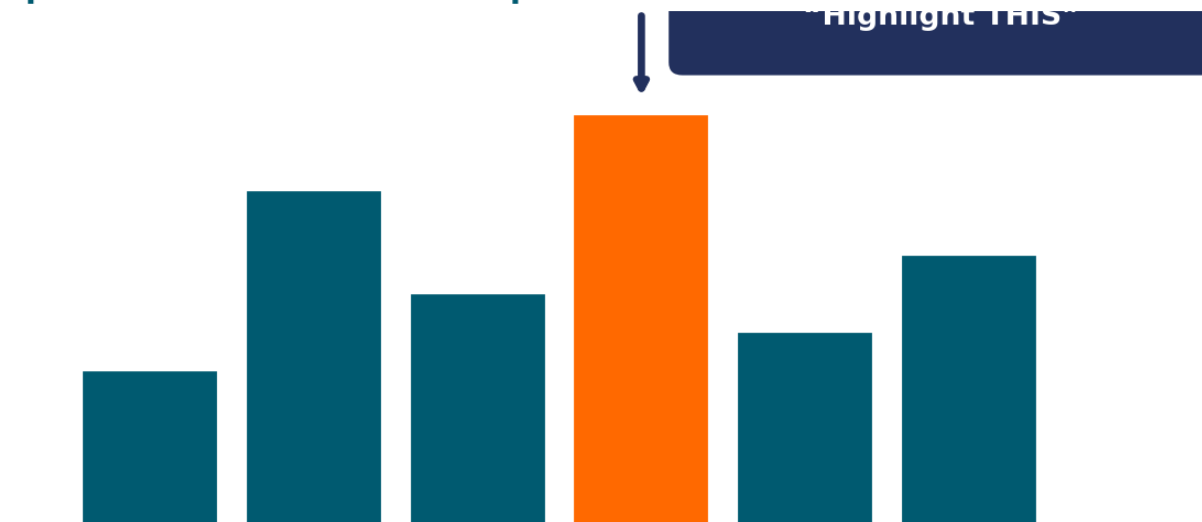
Brain fuses visual + audio speech.
 When they disagree, perception
 invents a third reading.
 Misaligned viz channels do the same.

DESIGN RULE

Predict user action to mask latency
 (Madhavan 2021). Fake it faster
 than you can render it.

Deixis in Viz — Pointing Resolves the Referent

Speech + Touch — deictic expressions bind words to referents



Language describes categories. Pointing picks instances. Every dashboard should accept both.

BOLT (1980)

"Put that there."

Voice + pointing. 45 years old,
still the founding demo of
multimodal viz interaction.

PRINCIPLE

Language is SYMBOLIC.

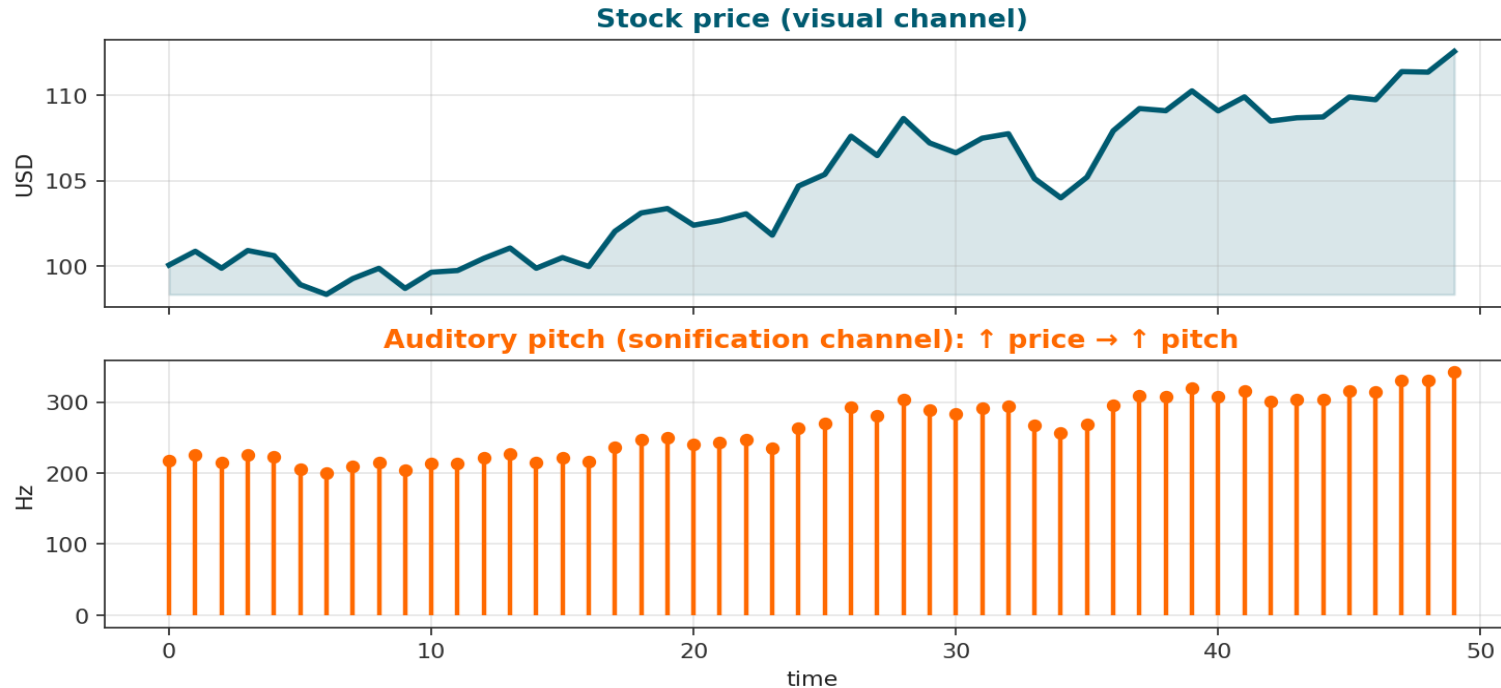
Gesture is INDEXICAL.

Combined, ambiguity collapses —
"this bar" needs both channels.

CASE

"Show me THIS metric for
THESE regions" — over a touch table.
Faster than any menu path.

Sonification — Sound as a Viz Channel



JavaScript

```
# Tone.js — encode a timeseries as a melody
const synth = new Tone.Synth().toDestination();
values.forEach((v, i) => {
  const pitch = 220 + (v - min) / (max - min) * 660;
  synth.triggerAttackRelease(pitch, '8n', i * 0.25);
})
```

ICAD TRADITION

30 years of sonification research.
Kramer (1994) handbook is the canonical reference. Sound is a visualization channel, not a sound effect.

ENCODINGS

Pitch $\propto$ value.
Timbre $\propto$ category.
Rhythm $\propto$ frequency.
Stereo pan $\propto$ position.

DESIGN RULE

Sonification needs a KEY —
a reference tone like a legend.
Without it, readers have no anchor.

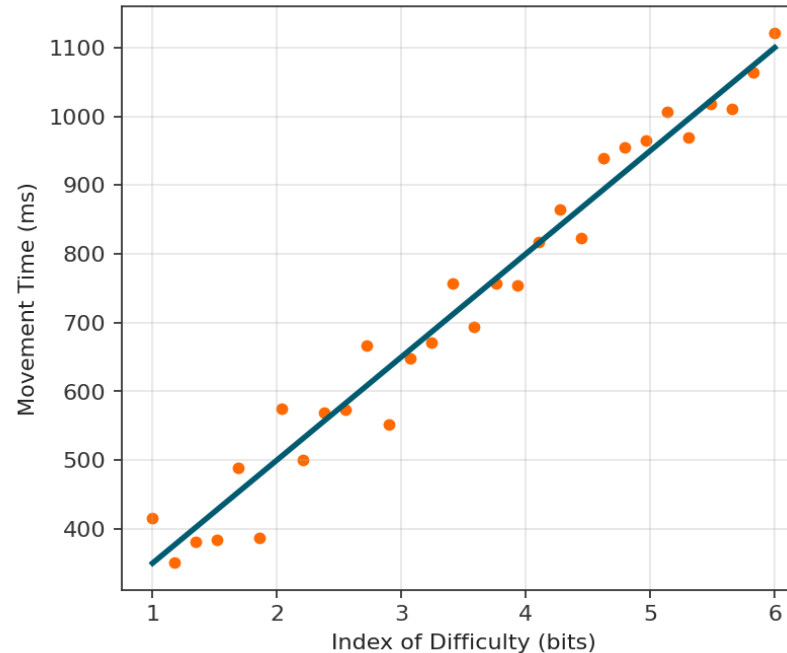
Fitts's Law — Affordance Design for Viz Controls

Fitts's Law in 3D Space



$$ID = \log_2\left(\frac{2D}{W}\right), \quad MT = a + b \cdot ID$$

MT scales linearly with ID



FITTS (1954)

$$MT = a + b \cdot \log_2(2D/W).$$

Predicts time → predicts friction.

Friction in viz controls is why readers stop exploring.

CONTROL SIZING

Slider, brush, lasso — size them by frequency of use.

Rare controls can be small; frequent controls must be big and close.

3D ADAPTATION

Place frequent controls at arm reach.

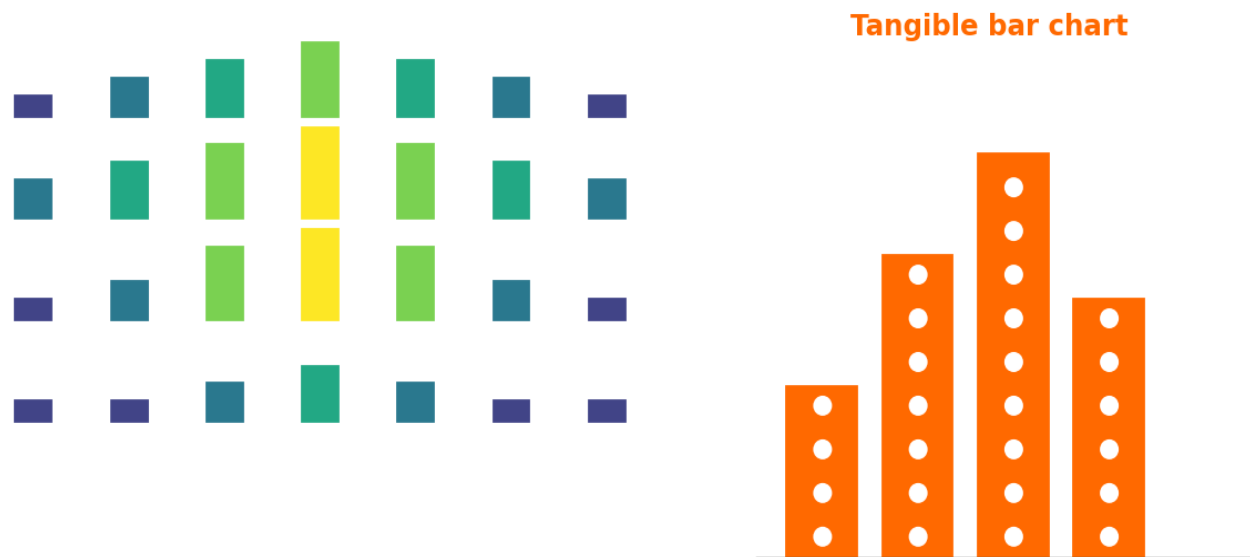
Rare ones past it. Distance IS the cost function.

$$ID = \log_2(2D / W) \quad MT = a + b \cdot ID$$

movement time is the cost of every interaction in your dashboard — budget it

Data Physicalization — Tangible Viz

Tangible User Interfaces (TUI) — data you can touch



Tangible bar chart

inFORM — dynamic shape display (MIT Tangible Media)

"When the chart is a thing, the thing is the chart."

SUBFIELD

Jansen & Dragicevic (2013):
 "Data Physicalization."
 Physical objects ARE the chart.
 Data you can hold and pass around.

EXAMPLES

Dear Data (Lupi & Posavec 2016).
 3D-printed population density maps.
 Climate ribbons (knitted weather).
 Ishii's Tangible Bits (1997).

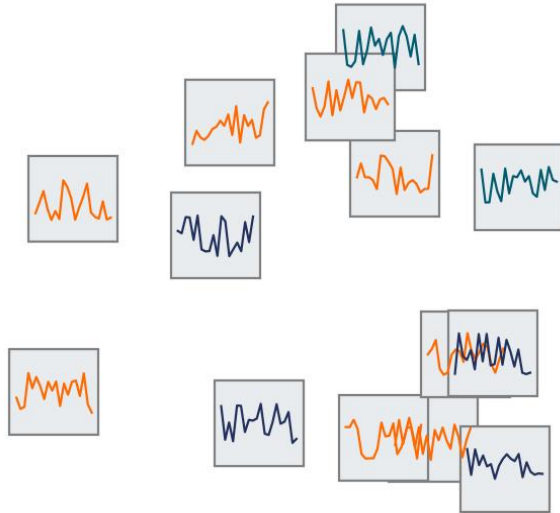
DESIGN BENEFIT

Physical affordances come free —
 you don't teach people to pick
 up a block. Rotation, grouping,
 proximity — all zero-learning idioms.

Affective Viz — Emotion as Data and Channel

User: high stress → simplified dashboard

Normal state: 14 panels



Detected stress → core KPI only



Heart-rate ↑ 28% → declutter automatically

Data visualization is not neutral. Every color choice has a feeling — own it.

EMOTION AS DATA

Visualize mood over time.
Sentiment across a corpus.
Viz subject: the inner state,
not the external world.

EMOTION AS CHANNEL

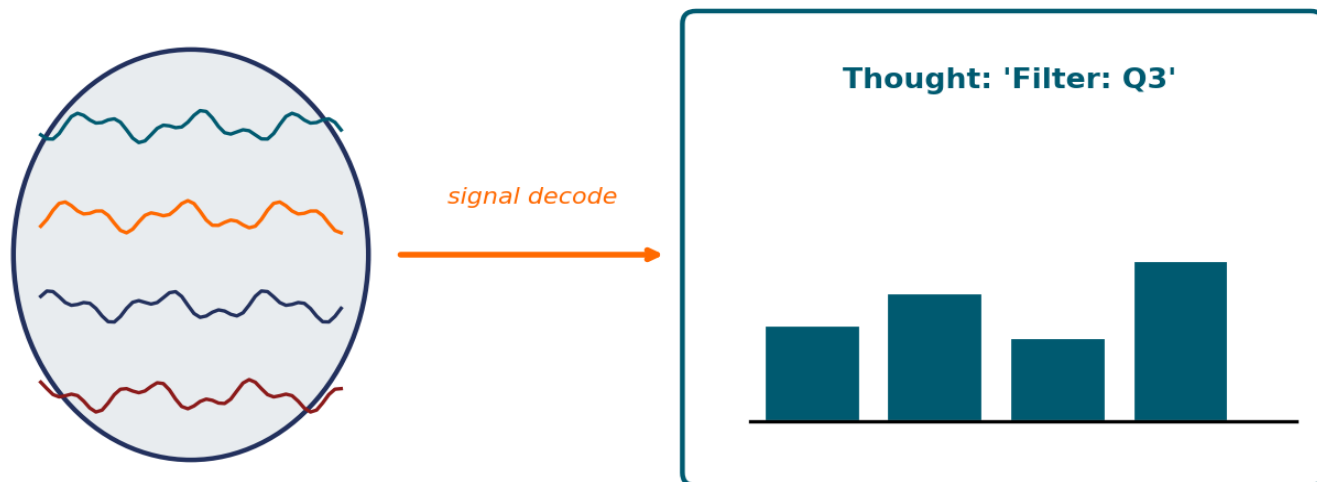
Viz that EVOKEs emotion —
Kosara's pragmatic/pragmatic axes.
Narrative viz (NYT, Pitch Interactive)
weaponizes color, pacing, scale.

ETHICS

Barrett (2019): emotion recognition
is cross-culturally contested.
Inferred affect must not masquerade
as ground truth.

Thought-Driven Viz — The BCI Horizon

Brain-Computer Interfaces: the ultimate multimodal endpoint



"The interaction bandwidth ceiling has been the bottleneck. BCI removes it — eventually."

EMERGING CHANNEL

BCI signals as VIZ INPUT.

Imagined commands filter a chart.

Motor imagery selects a region.

Neural signal, same viz grammar.

ADAPTIVE VIZ

Cognitive load sensing → simplify the view when readers get stressed.

The chart responds to the reader's state, not only their input.

HORIZON

Willett et al. (Nature 2023):

62 wpm decoded speech from ECoG.

Consumer BCI for viz is a decade out — but the design space opens now.